

拓扑缺陷网络中的连续对称性反常

Qiang Jia¹、Ran Luo²、Jiahua Tian³、Yi-Nan Wang^{2,4,5} 和 Yi Zhang⁶

¹ 韩国科学技术院物理系, 大田34141, 韩国 ² 北京大学物理学院

³ 华东师范大学物理与电子科学学院, 中国上海, 200241 ⁴ 北京大学高能物理中心

⁵ 安徽合肥230026, 彭桓武基础理论研究中心, 以及 ⁶ Kavli IPMU (WPI) , UTIAS, 东京大学, 日本千叶县柏市277-8583 (日期: 2025年12月22日)

我们证明, 在一般时空维数中, 实现全局对称性的拓扑缺陷网的重排, 可以检测具有连续味对称性的量子场论的 ’t Hooft 反常; 在二维情形中, 这具体体现为缺陷线的 F 变换。通过将缺陷对偶为平坦背景规范场构型, 我们利用对称群的各种上同调数据刻画 ’t Hooft 反常, 其中具有离散拓扑的李群的上同调发挥核心作用。我们发现, 在平坦规范化的情形下, 作为 ’t Hooft 反常数学描述的自然结果, 额外维度会自然涌现。

引言。若对量子场论 (QFT) 的全局对称性进行规范化会导致不一致性, 即存在将其提升为规范对称性的障碍 $e^{2\pi i\mathcal{A}(A,g)}$, 则称该全局对称性具有 ’t Hooft 反常。从紫外到红外, ’t Hooft 反常具有稳健性, 这使其成为研究量子场论性质 (即使在强耦合区域) 的特别有效工具[2,3]。’t Hooft 反常表现为规范变换下 $Z[A]$ 获得的非平凡相位因子 $A \rightarrow g^{-1}Ag + g^{-1}dg$, 其中 $Z[A]$ 是与背景场 A 耦合的量子场论的配分函数, 而对于 $\mathcal{A}(A,g)$, 该相位无法由任何由 A 构成的局域反项消除。传统上, 对于连续 G , 可利用小规范变换的下降方程对其进行计算[4-8]; 更一般地, 当 g 与单位元不连通时, 已证明它由 eta 不变量给出9-12。

近年来, 凝聚态物理和高能物理领域都重新兴起了对 ’t Hooft 反常的研究[13-17]。得益于将对称性作用重新表述为时空中对称性算符与带电缺陷的链接 $H^{d+1}(G,U(1))$, 如今有限群对称性的作用可以与连续对称性置于同等地位进行研究。然而, 有限群对称性的 ’t Hooft 反常可以直接计算为缺陷网重排所对应的 F 符号, 这些符号通常由群上同调的元素描述 (例如, 在 d 时空维数中, 0-形式对称群 G 对应的 $H^{d+1}(G,U(1))$) [24-27]; 相比之下, 用于计算平坦规范化下连续群对称性 ’t Hooft 反常的类似算法长期以来仍难以寻得。

在本文中, 我们提出了这样一种适用于连续对称性的算法, 从而填补缺失的一环, 使有限对称性与连续对称性的处理能够完全并行。我们首先证明, 缺陷网可以对偶为平坦背景场构型。这实际上确立了插入缺陷与将守恒流耦合到平坦背景场之间的等价性。通过对偶化缺陷

具体而言, 我们将以完全一般的方式展示, ’t Hooft 异常如何表现为拓扑缺陷按照文献 21 所暗示的方式重新排列时产生的相位因子。随后, 我们通过二维量子场论中拓扑缺陷线的 F-move [16, 17, 28-30] 具体计算该异常, 并表明该异常由 G 的上同调数据刻画。

非常重要的是, 不同于有限群 G 的情形——在那里群上同调 $H^3(G,U(1)) \cong H^3(BG,U(1))$ 有明确无歧义的定义——对于李群上同调的精确定义必须谨慎处理。我们强调, 对于李群 G , 正确的版本应为 $H^3(G^\delta,U(1))$; 其定义是将 G 赋予离散拓扑后得到的群上同调 [31]。尽管 $H^3(G^\delta,U(1))$ 是一个极其复杂的数学对象, 但存在从群 $H^4(BG,\mathbb{Z})$ (G 的异常多项式所在之处) 到 $H^3(G^\delta,U(1))$ 的嵌入映射。因此, 通常的异常多项式 $\sim F \wedge F$ 可以优雅地嵌入这一统一框架中。

我们发现, 支撑这些关于 G 的不同上同调群的统一数学结构表明, 具有 ’t Hooft 异常的量子场论自然地位于高一维异常理论的边界上。在我们的形式体系中, 将上述对 ’t Hooft 异常的上同调刻画推广到更高维时空是直接的。

作为详细例子, 我们分别讨论简单 G 和 $U(1)$ 的情形。对于紧致、单连通的简单李群 G , $3d$ 体异常理论是由称为 level 的整数 $k \in \mathbb{Z}$ 刻画的非阿贝尔 Chern-Simons 不变量。它具有局部表达式, 即 Chern-Simons 3-形式 $I_3 = CS_k[A] = \frac{k}{2}\text{tr}(A \wedge F - \frac{1}{3}A \wedge A \wedge A)$ 。我们明确地将拓扑缺陷网络沿 F-move 提升到三维体内, 并证明在赋予 $F = 0$ 后, $CS_k[A]$ 中的三次项恰好生成 G 的 de Rham 上同调群 $H_{\text{dR}}^3(G,\mathbb{R})$ 。

对于诸如 $G = U(1)$ 这样的非单连通李群, 我们指出, 场强 $F = dA$ 在此情形下不会

Anomaly of Continuous Symmetries from Topological Defect Network

Qiang Jia¹, Ran Luo², Jiahua Tian³, Yi-Nan Wang^{2,4,5}, and Yi Zhang⁶

¹*Department of Physics, Korea Advanced Institute of Science & Technology, Daejeon 34141, Korea*

²*School of Physics, Peking University*

³*School of Physics and Electronic Science, East China Normal University, Shanghai, China, 200241*

⁴*Center for High Energy Physics, Peking University*

⁵*Peng Huanwu Center for Fundamental Theory, Hefei, Anhui 230026, China and*

⁶*Kavli IPMU (WPI), UTIAS, The University of Tokyo, Kashiwa, Chiba 277-8583, Japan*

(Dated: December 22, 2025)

We show that the ’t Hooft anomaly of a quantum field theory with continuous flavor symmetry can be detected from rearrangements of the topological defect webs implementing the global symmetry in general spacetime dimension, which is concretized in 2D by the *F-moves* of the defect lines. Via dualizing the defects to flat background gauge field configurations, we characterize the ’t Hooft anomaly by various cohomological data of the symmetry group, where the cohomology of Lie groups with discrete topology plays the central role. We find that an extra dimension emerges naturally as a consequence of the mathematical description of the ’t Hooft anomaly in the case of flat gauging.

Introduction. A global symmetry of a Quantum Field Theory (QFT) is said to have a ’t Hooft anomaly if gauging it leads to an inconsistency, i.e. an obstruction to promoting it to a gauge symmetry [1]. The robustness of ’t Hooft anomaly across ultraviolet to infrared makes it a particularly effective tool to study the properties of QFT even in the strongly coupled region [2, 3]. The ’t Hooft anomaly manifests itself as a nontrivial phase factor $e^{2\pi i\mathcal{A}(A,g)}$ acquired by $Z[A]$ under gauge transformation $A \rightarrow g^{-1}Ag + g^{-1}dg$, where $Z[A]$ is the partition function of the QFT coupled to background field A and for $\mathcal{A}(A,g)$ that cannot be cancelled by any local counterterm built out of A . Traditionally for continuous G , it can be evaluated using the *descent equations* for small gauge transformations [4–8] and more generally shown to be the the *eta-invariant* when g is disconnected from the identity [9–12].

Recent years have witnessed a revival of the study of ’t Hooft anomalies in both condensed matter and high-energy physics [13–17]. Thanks to the reformulation of the symmetry action as linking symmetry operator and charged defect in spacetime [18–23], the action of finite group symmetries is now conveniently studied on equal footing with that of continuous symmetries. However, whereas the ’t Hooft anomaly of discrete group symmetry can be computed directly as the *F-symbol* associated with rearrangements of webs of defects, which are often described by elements of the group cohomology (e.g. $H^{d+1}(G,U(1))$ for a 0-form symmetry group G in d -spacetime dimensions)[24–27], an analogous algorithm to compute the ’t Hooft anomaly for continuous group symmetries *in flat gauging* has long remained elusive.

In this work, we present such an algorithm for continuous symmetries, thereby filling the missing corner to place the treatment of finite and continuous symmetries on a completely parallel footing. We first show that a web of defects can be dualized to be a flat background field configuration. This effectively establishes the equivalence between the insertion of defects and coupling the conserved current to flat background fields. Via the du-

alization, we show in full generality how the ’t Hooft anomaly manifests itself as the phase factor arising from rearrangements of the topological defects in the manner hinted in [21]. We then compute the anomaly concretely via *F-moves* of the *topological defect lines* in 2D QFT [16, 17, 28–30] and show that the anomaly is characterized by cohomological data of G .

Very importantly, unlike the case of a finite group G where the group cohomology $H^3(G,U(1)) \cong H^3(BG,U(1))$ is unambiguously defined, one should be careful about the exact definition for Lie group cohomology. We emphasize that for a Lie group G the correct version should be $H^3(G^\delta,U(1))$, which is defined to be the group cohomology when G is equipped with discrete topology [31]. Despite that $H^3(G^\delta,U(1))$ is a highly complicated mathematical object, there exists an injection map from the group $H^4(BG,\mathbb{Z})$, where the anomaly polynomial for G lives, to $H^3(G^\delta,U(1))$. Thus the usual anomaly polynomial $\sim F \wedge F$ can be elegantly embedded in this unified framework.

We find that the unifying mathematical structure underlying these various cohomology groups of G suggests that the QFT with a ’t Hooft anomaly naturally lives on the boundary of an *anomaly theory* in one higher dimension. The generalization of the above cohomological characterization of ’t Hooft anomaly to higher-dimensional spacetime is immediate in our formalism.

For detailed examples, we discuss the cases of a simple G and $U(1)$ separately. For a compact simply connected and simple Lie group G , the $3d$ bulk anomaly theory is the non-abelian Chern–Simons invariant characterized by an integer $k \in \mathbb{Z}$ called the *level*. It has a local expression as the Chern–Simons 3-form $I_3 = CS_k[A] = \frac{k}{2}\text{tr}(A \wedge F - \frac{1}{3}A \wedge A \wedge A)$. We explicitly uplift the topological defect networks across the F-move to a three-dimensional bulk, and show that after assigning $F = 0$, the cubic term in $CS_k[A]$ precisely generates the de Rham cohomology group $H_{\text{dR}}^3(G,\mathbb{R})$ for G .

For a non-simply connected Lie group such as $G = U(1)$, we point out that the field strength $F = dA$ would

在结点处消失。这与平坦规范化的要求并不矛盾，因为平坦规范化的精确定义应是：绕拓扑平凡回路 $\mathcal{C}$ 的 holonomy $\exp(\oint_{\mathcal{C}} A)$ 为平凡。这样的非零 F 正是 F-move 前后非平凡 $U(1)$ 相位因子的来源；我们通过在二维时空上对 ϵF 积分来明确计算该相位因子，其中 ϵ 是生成 F-move 的有限规范变换参数。相位因子 32

$$\omega(e^{i\alpha_1}, e^{i\alpha_2}, e^{i\alpha_3}) = e^{i\alpha_1(\alpha_2 + \alpha_3 - [\alpha_2 + \alpha_3])/(2\pi)} \quad (1)$$

其中 $\alpha_i \in [0, 2\pi)$ 和 $[r]$ 表示超过 2π 的 r 部分，与 $H^3(\mathbb{Z}_N, U(1))$ 生成元的 $N \rightarrow \infty$ 极限一致。

我们还想评论的是，(1) 与新近提出的连续对称性的范畴表述 [33] 完全一致。也就是说，如果假定对称性范畴 $\mathbf{Vec}_G^\omega$ 和 Drinfeld 中心 $Z(\mathbf{Vec}_G^\omega)$ 的定义可以推广到 $G = U(1)$ 的情形，那么所有范畴论数据都与 [34] 中的数学结果一致，而这些结果已在 [33] 中得到确认。

我们的结果适用于所有紧致连通李群。根据结构定理 [35]，紧致连通李群被环面与单非阿贝尔群的直积有限覆盖，因此其分类可归结为单群的分类。在此背景下，限制于紧致连通单李群 G 或 $U(1)$ 已经足够。

最后，尽管具体例子针对 $d = 2$ 情形进行讨论，我们的框架同样适用于一般 d 维时空中的连续 0-形式对称性。

缺陷网的对偶化。(拉格朗日) QFT $\mathcal{T}_G$ 在 d 维时空 M_d 中的拓扑缺陷网 $\mathcal{W}$ 的真空期望值可计算为：

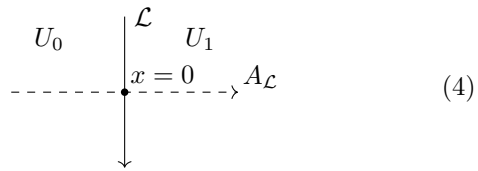
$$\langle \mathcal{W} \rangle := \int \mathcal{D}\psi \, e^{iS[\psi] + \int_{M_d} \text{tr}(A_{\mathcal{W}} \wedge \star J)}, \quad (2)$$

其中 ψ 表示 $\mathcal{T}_G$ 的局域算符集合，而 $J(\psi)$ 是连续全局 G -对称性的守恒流。 $A_{\mathcal{W}}$ 代表与缺陷网 $\mathcal{W}$ 对应的普通平直规范场。为说明对偶化过程，我们考虑 $\mathbb{R}^2$ 中沿 $x = 0$ 、标记为 $g_\alpha = e^{i\alpha} \in G$ 的缺陷线 $\mathcal{L}$ ，这意味着一个携带全局 G -对称性电荷的粒子从左向右穿过 $x = 0$ 时，在 g_α 的表示下发生变换，见 4。我们将 $\mathcal{L}$ 的对偶化 $A_{\mathcal{L}}$ 定义为满足以下条件的背景规范场：对于任意仅穿过 $x = 0$ 一次的路径 ℓ ，均有 $\mathcal{P}e^{\int_{\ell} A_{\mathcal{L}}} = e^{i\alpha}$ ，这反过来意味着 $A_{\mathcal{L}}$ 必须是平直的。在此条件下，我们可以写出：

$$A_{\mathcal{L}} = i\alpha\delta(x)dx = e^{-i\alpha H(x)} de^{i\alpha H(x)} \quad (3)$$

其中 $H(x)$ 是 Heaviside 函数，并且确实可以验证 $dA_{\mathcal{L}} + A_{\mathcal{L}} \wedge A_{\mathcal{L}} = 0$ 。等价地， $\mathcal{L}$ 定义了

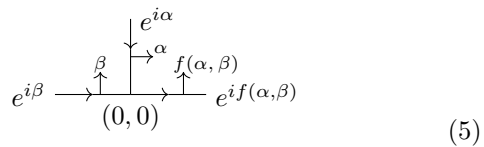
在 $\mathbb{R}^2$ 的以下开覆盖上，为 $g_0 = e$ 和 $g_1 = e^{i\alpha}$ 配置规范场 $A_{\mathcal{L}}|_{U_i} = g_i^{-1} dg_i$ ：



$$\begin{array}{c} U_0 \quad \quad U_1 \\ \text{-----} \quad \quad \text{-----} \quad \rightarrow A_{\mathcal{L}} \\ \quad \quad \quad \downarrow \\ \quad \quad \quad x=0 \end{array} \quad (4)$$

其中，两个斑块 $U_0 = (-\infty, \epsilon)$ 和 $U_1 = (-\epsilon, \infty)$ 沿 $(-\epsilon, \epsilon)$ 粘合，这里 ϵ 为小的正数。这里我们约定，相对于从 U_0 到 U_1 的规范变换方向，缺陷线的方向按右手方向旋转。显然，上述讨论可以推广到一般 d 维中的余维一 $\mathcal{W}$ 网络。

任何缺陷网络（由线构成）的基本构件是三价连接点：



$$\begin{array}{c} \downarrow e^{i\alpha} \\ \nearrow \beta \quad \searrow \alpha \\ e^{i\beta} \rightarrow \quad \rightarrow \quad \rightarrow e^{if(\alpha, \beta)} \\ (0,0) \end{array} \quad (5)$$

对应于群元素 $e^{i\alpha}$ 的背景规范场 $A_{\mathcal{L}}$ 也记作 α 。对于在 $\mathbb{R}^2$ 上定义的全局 g ，当 G 具有平凡的 $\pi_1(G)$ 时，这个三价连接点可以涂抹开而得到带有 $F = dA + A \wedge A = 0$ 的 $A = g^{-1}dg$ ，因为群流形上从单位元到 $e^{i\alpha}e^{i\beta}$ 的所有路径都是同伦的。然而，当 G 具有非平凡的 $\pi_1(G)$ 时，例如 $G = U(1)$ ，任何形式的涂抹都无法得到全局定义的 g ；并且存在具有非平凡 $F = dA$ 的 $U(1)$ 规范场的平坦构型。这一微妙之处对于推导阿贝尔群的反常至关重要。

现在我们澄清连续 G 全局对称性的平坦规范化概念，它由以下两个等价条件给出：

1. 平坦规范化等价于在新的配分函数中对所有可能插入的拓扑缺陷网络求和。
2. 对规范场 A 的构型求和，其中规范场绕拓扑平凡回路 $\mathcal{P} \exp(\oint_{\mathcal{C}} A) = e$ 的 holonomy 为 G 的单位元。

来自 F-变换与李群上同调的 't Hooft 异常。定义连续 G 的对偶化后，可以分别将 $\mathcal{T}_G$ 耦合到平坦的 $A_{\mathcal{W}}$ - 和 $A_{\mathcal{W}'}$ 背景，从而比较 $\langle \mathcal{W} \rangle$ 与 $\langle \mathcal{W}' \rangle$ 。设 $A_{\mathcal{W}'} = e^{-i\Lambda} A_{\mathcal{W}} e^{i\Lambda} + e^{-i\Lambda} de^{i\Lambda}$ 对于由 Λ 参数化的规范变换 $\psi \rightarrow e^{i\Lambda} \psi$ ，则有：

$$\langle \mathcal{W}' \rangle = \int \mathcal{D}\psi \, e^{iS[\psi] + \int_{M_d} \text{tr}(A_{\mathcal{W}'} \wedge \star J)} = e^{2\pi i A[A_{\mathcal{W}}; \Lambda]} \langle \mathcal{W} \rangle \quad (6)$$

not vanish at the junction point for this case. This is not inconsistent with the requirement of flat gauging, as the precise definition of flat gauging should be the triviality of the holonomy $\exp(\oint_{\mathcal{C}} A)$ around a topologically trivial loop $\mathcal{C}$. Such a non-vanishing F is precisely the source of the non-trivial $U(1)$ phase factor across the F-move, which we compute explicitly from the integration of ϵF over the 2D spacetime, where ϵ is the finite gauge transformation parameter generating the F-move. The phase factor [32]

$$\omega(e^{i\alpha_1}, e^{i\alpha_2}, e^{i\alpha_3}) = e^{i\alpha_1(\alpha_2 + \alpha_3 - [\alpha_2 + \alpha_3])/(2\pi)} \quad (1)$$

with $\alpha_i \in [0, 2\pi)$ and $[r]$ denoting the part of r exceeding 2π , is consistent with the $N \rightarrow \infty$ limit of the generator of $H^3(\mathbb{Z}_N, U(1))$.

We would also like to comment that (1) is perfectly consistent with the newly proposed categorical formulation of continuous symmetry [33]. Namely, if one pretends that the definition of the symmetry category $\mathbf{Vec}_G^\omega$ and Drinfeld center $Z(\mathbf{Vec}_G^\omega)$ can be generalized to the case of $G = U(1)$, all the category theoretical data are consistent with the mathematical results in [34], which are confirmed in [33].

Our results can be applied to all compact connected Lie groups. From the structure theorem [35], compact connected Lie groups are finitely covered by direct products of tori and simple non-abelian groups, such that their classification reduces to that of simple groups. The restriction to a compact connected and simple Lie group G or $U(1)$ is sufficient in this context.

Finally, although the detailed examples are discussed for the $d = 2$ case, our framework applies to continuous 0-form symmetries in general d spacetime dimensions as well.

Dualization of the defect web. The VEV of a topological defect web $\mathcal{W}$ of a (Lagrangian) QFT $\mathcal{T}_G$ in d -dimensional spacetime M_d can be computed as:

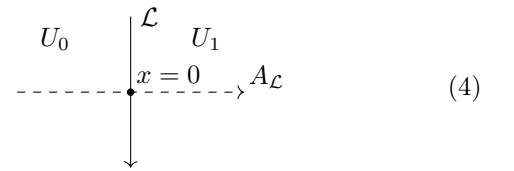
$$\langle \mathcal{W} \rangle := \int \mathcal{D}\psi \, e^{iS[\psi] + \int_{M_d} \text{tr}(A_{\mathcal{W}} \wedge \star J)}, \quad (2)$$

where ψ stands for the set of local operators of $\mathcal{T}_G$ and $J(\psi)$ is the conserved current of the continuous global G -symmetry. $A_{\mathcal{W}}$ standards for the ordinary flat gauge field corresponding to the defect web $\mathcal{W}$. To illustrate the dualization, we consider a defect line $\mathcal{L}$ labeled by $g_\alpha = e^{i\alpha} \in G$ along $x = 0$ in $\mathbb{R}^2$, meaning that a particle charged under the global G -symmetry is transformed under a representation of g_α when crossing $x = 0$ from left to right, see (4). We define the dualization $A_{\mathcal{L}}$ of $\mathcal{L}$ to be the background gauge field such that $\mathcal{P}e^{\int_{\ell} A_{\mathcal{L}}} = e^{i\alpha}$ for an arbitrary path ℓ crossing $x = 0$ once, which in turn means that $A_{\mathcal{L}}$ must be *flat*. Given the condition, we can write:

$$A_{\mathcal{L}} = i\alpha\delta(x)dx = e^{-i\alpha H(x)} de^{i\alpha H(x)} \quad (3)$$

where $H(x)$ is the Heaviside function, and we can indeed check that $dA_{\mathcal{L}} + A_{\mathcal{L}} \wedge A_{\mathcal{L}} = 0$. Equivalently, $\mathcal{L}$ defines

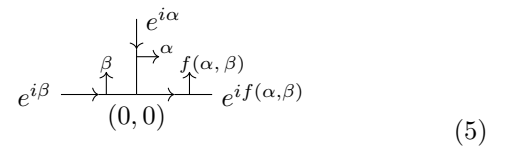
the gauge configuration $A_{\mathcal{L}}|_{U_i} = g_i^{-1} dg_i$ for $g_0 = e$ and $g_1 = e^{i\alpha}$ on the following open cover of $\mathbb{R}^2$:



$$\begin{array}{c} U_0 \quad \quad U_1 \\ \text{-----} \quad \quad \text{-----} \quad \rightarrow A_{\mathcal{L}} \\ \quad \quad \quad \downarrow \\ \quad \quad \quad x=0 \end{array} \quad (4)$$

where the two patches $U_0 = (-\infty, \epsilon)$ and $U_1 = (-\epsilon, \infty)$ are glued along $(-\epsilon, \epsilon)$ for small positive ϵ . Here we adopt the convention that the direction of the defect line rotates right-handedly with respect to the direction of the gauge transformation from U_0 to U_1 . Clearly, the above discussions can be generalized to a codimension-one web of $\mathcal{W}$ in general d dimensions.

The fundamental building block of any defect web (of lines) is the trivalent junction:



$$\begin{array}{c} \downarrow e^{i\alpha} \\ \nearrow \beta \quad \searrow \alpha \\ e^{i\beta} \rightarrow \quad \rightarrow \quad \rightarrow e^{if(\alpha, \beta)} \\ (0,0) \end{array} \quad (5)$$

The background gauge field $A_{\mathcal{L}}$ corresponding to the group element $e^{i\alpha}$ is also labeled, with α . This trivalent junction can be smeared out to give $A = g^{-1}dg$ with $F = dA + A \wedge A = 0$ for a globally defined g on $\mathbb{R}^2$ for G with trivial $\pi_1(G)$, since all paths from identity to $e^{i\alpha}e^{i\beta}$ on the group manifold are homotopic. However, a globally defined g cannot be obtained by any kind of smearing for G with non-trivial $\pi_1(G)$, e.g. for $G = U(1)$, and there exist flat configurations of $U(1)$ gauge fields with non-trivial $F = dA$. This subtlety is crucial for the derivation of anomalies for abelian groups.

Now we clarify the notion of flat gauging of continuous G global symmetry, given by the following two equivalent conditions:

1. A flat gauging is equivalent to summing over all possible insertions of topological defect networks in the new partition function.
2. Summing over configurations of gauge fields A , where the holonomy of the gauge field around a topologically trivial loop $\mathcal{P} \exp(\oint_{\mathcal{C}} A) = e$, the identity element of G .

't Hooft anomaly from F-move and the Lie group cohomology. Having defined the dualization for continuous G , one can compare $\langle \mathcal{W} \rangle$ with $\langle \mathcal{W}' \rangle$ by coupling $\mathcal{T}_G$ to flat $A_{\mathcal{W}}$ - and $A_{\mathcal{W}'}$ -backgrounds, respectively. Suppose $A_{\mathcal{W}'} = e^{-i\Lambda} A_{\mathcal{W}} e^{i\Lambda} + e^{-i\Lambda} de^{i\Lambda}$ for gauge transformation $\psi \rightarrow e^{i\Lambda} \psi$ parametrized by Λ , we have:

$$\langle \mathcal{W}' \rangle = \int \mathcal{D}\psi \, e^{iS[\psi] + \int_{M_d} \text{tr}(A_{\mathcal{W}'} \wedge \star J)} = e^{2\pi i A[A_{\mathcal{W}}; \Lambda]} \langle \mathcal{W} \rangle \quad (6)$$

其中 $\mathcal{A}[A_W; \Lambda] \neq 0$ 是量子反常。因此，基于我们的对偶化，由重新排列缺陷网络所产生的反常计算，归结为寻找连接这两个对偶平坦场构型的规范变换。

提醒一下，二维玻色理论的 $\mathbb{Z}_m$ -对称性的 't Hooft anomaly 众所周知由 $H^3(\mathbb{Z}_m, U(1))$ 刻画，即 $\mathbb{Z}_m$ 的 3rd 群上调群，取 $U(1)$ 为系数，其中缺陷网图1中F-move的相位因子 $e^{2\pi i a_1(a_2+a_3-a_2+a_3)/m}$ 属于该上调群[28]。在d维时空中，

图1. 二维中 $G = \mathbb{Z}_m$ 0-形式对称群的缺陷网F-move。 $a_i \in \{0, 1, \dots, m-1\}$ 和 $\bar{m}$ 定义为 $m \bmod \mathbb{Z}$ 。

其刻画由 $H^{d+1}(G, U(1))$ 给出[16,26]。为填补连续 G 的类似刻画这一空白，我们将关注所感兴趣的连续 G ，并在平直规范化的情形下计算其 anomalies，即将 $\mathcal{T}_G$ 耦合到平直背景场；这最接近离散情形，因为其中所有背景场自动平直。

我们证明，'t Hooft anomaly 编码在 $\langle \mathcal{M} \rangle = e^{2\pi i \mathcal{A}[A_\Delta; \Lambda]} \langle \mathcal{M} \rangle$ 中的相位因子内（参见式(6)），该相位因子对应于位于 M_2 上的 $\mathcal{A}$ 和 $\mathcal{A}$ 缺陷网，它们处于具有李群 G 的 $\mathcal{T}_G$ 中。这里 Λ 是从 A_Δ 到 A_Δ 的规范变换。

用现代语言来说[12]， $\mathcal{A}[A_\Delta; \Lambda]$ 可计算为（参见附录E中的式E9）：

$$\mathcal{A}(A_\Delta; \Lambda) = \int_{M_3} CS_k[A(t)], \quad (7)$$

利用等级为 k 的陈-西蒙斯不变量 CS_k ，其中 $t \in [0, 1]$ 参数化规范场构型空间中的一个映射柱面，沿该柱面 $A(t) := g(t)^{-1} A_\Delta g(t) + g(t)^{-1} dg(t)$ 对于 $g(x, y, t=0) = 1 \in G$ 和 $g(x, y, t=1) = e^{i\Lambda(x, y)}$ 从 A_Δ 插值到 A_Δ 。我们采用归一化 $CS_k[A] := \frac{k}{2} \text{tr}(A \wedge dA + \frac{2}{3} A \wedge A \wedge A)$ 。

通过考察陈-西蒙斯不变量的局部表达式，我们可以得到反常的其他一些表现形式。在本讨论中，由于 G 是单连通的，我们可以为平坦规范场指定 $F = 0$ 。在 M_3 体内，我们的规范场 $A(t)$ 是平坦的。在局部片 U_i 上，我们有

在 M_3 的一个局部片上有 $A \wedge A = \frac{k}{2} \text{tr}(A \wedge F - \frac{1}{3} A \wedge A \wedge A)$ 。为简单起见，我们将研究范围限制在这样一类理论：其反常计算中使用的陈-西蒙斯不变量源自 $H^4(BG, \mathbb{Z})$ 中的整类（见附录C中的公式C11）。该条件限制了允许的最小层级 k ，其取值取决于 G 。我们逐一讨论这些情形，并为简洁起见省略下标 k 36。

从 A 到 A 的插值示于图2最左侧的图中。不难看出，按照我们的约定，前面的构型等价于 A ，而后面的构型等价于 A 。由于 $\mathcal{A}[A_\Delta; \Lambda]$ 显然不依赖于映射圆柱的小变形，即不依赖于 g 的细节，只要边界构型 A 和 A 固定，我们便进一步将 A 的顶点与 Δ 中具有相同标签的顶点，在两份 $M_2 \times \{0\}$ 和 $M_2 \times \{1\}$ 上分别粘合，从而得到图2中间的构型。这就是相应缺陷网之间的差异；中间的构型还可以进一步变形为图2最右侧所示的带装饰四面体，其中缺陷和规范变换的方向由我们的约定（4）确定。这个过程可以理解为反常流入——与有限群情形[16]十分类似，即可以在体内附加一个额外的3-单形来抵消反常，而体内四面体的装饰描述了 BG^δ 中3-单形 $\langle e, g_1, (g_1 g_2), (g_1 g_2 g_3) \rangle$ 到体内的拉回。这里 G^δ 是带离散拓扑的李群 G ，而 BG^δ 是分类平坦 G -丛的 Eilenberg-MacLane 空间 $K(G, 1)$ ；我们将分类映射记为 $\varphi^\delta: M_3 \rightarrow BG^\delta$ 37。

与有限群情形类似，就分类空间而言，异常可通过在 $\langle e, g_1, (g_1 g_2), (g_1 g_2 g_3) \rangle$ 上评价上调调类 $[\Omega] \in H^3(BG^\delta, \mathbb{R}/\mathbb{Z})$ 得到。这里我们将系数 $U(1) = \mathbb{R}/\mathbb{Z}$ 视为加法群。用 Ω 与 φ^δ 作拉回并在三维时空上积分，我们得到Dijkgraaf-Witten相位

$$\mathcal{A}(A_\Delta; \Lambda) = \int_{M_3} (\varphi^\delta)^*(\Omega), \quad (8)$$

换言之，我们声称，通过F-move找到的 Ω 拉回为 Chern-Simons 不变量 CS 。证明见附录C中的式（8）。

对于某个 $\tilde{g}_i: U_i \rightarrow G$ ， $A_i := A|_{U_i} = \tilde{g}_i^{-1} d\tilde{g}_i$ ，以及 $\int_{U_i} -\frac{k}{6} \text{tr}(A_i \wedge A_i \wedge A_i) = \int_{U_i} k \tilde{g}_i^* \omega_3 = \int_{\tilde{g}_i(U_i)} k \omega_3$ (9)

其中 $\tilde{g}_i^*$ 是从 $\Omega^*(G)$ 到 $\Omega^*(U_i)$ 的拉回，并且

where $\mathcal{A}[A_W; \Lambda] \neq 0$ is the quantum anomaly. Therefore, given our dualization, the computation of anomaly arises from rearrangements of the defect webs amounts to finding the gauge transformation interpolating the dual flat field configurations.

As a reminder, the 't Hooft anomaly of $\mathbb{Z}_m$ -symmetry of a 2D bosonic theory is well-known to be characterized by $H^3(\mathbb{Z}_m, U(1))$, the 3rd group cohomology of $\mathbb{Z}_m$ with $U(1)$ -coefficient, in which the phase factor $e^{2\pi i a_1(a_2+a_3-a_2+a_3)/m}$ of the F-move of the defect webs in Figure 1 lives [28]. In d -spacetime dimensions the

FIG. 1. The F-move of defect web for $G = \mathbb{Z}_m$ 0-form symmetry group in 2D. $a_i \in \{0, 1, \dots, m-1\}$ and $\bar{m}$ is defined as $m \bmod \mathbb{Z}$.

characterization is given by $H^{d+1}(G, U(1))$ for discrete G [16, 26]. To fill the gap of a similar characterization for continuous G , we will focus on continuous G of interests, and compute their anomalies in the case of *flat gauging*, i.e. from coupling $\mathcal{T}_G$ to *flat* background field that is the closest cousin of the discrete case where all background fields are automatically flat.

We show that the 't Hooft anomaly is encoded in the phase factor in $\langle \text{triangle} \rangle = e^{2\pi i \mathcal{A}[A_\Delta; \Lambda]} \langle \text{triangle} \rangle$ (cf. (6)) for the defect webs triangle and triangle living on M_2 in $\mathcal{T}_G$ with Lie group G . Here Λ is the gauge transformation from A_Δ to A_Δ .

In the modern language [12], $\mathcal{A}[A_\Delta; \Lambda]$ can be calculated as (cf. Eq. (E9) in Appendix E):

$$\mathcal{A}(A_\Delta; \Lambda) = \int_{M_3} CS_k[A(t)], \quad (7)$$

using the level- k Chern-Simons invariant CS_k where $t \in [0, 1]$ parametrizing a mapping cylinder in the space of gauge field configurations along which $A(t) := g(t)^{-1} A_\Delta g(t) + g(t)^{-1} dg(t)$ for $g(x, y, t=0) = 1 \in G$ and $g(x, y, t=1) = e^{i\Lambda(x, y)}$ interpolates from A_Δ to A_Δ . We use the normalization $CS_k[A] := \frac{k}{2} \text{tr}(A \wedge dA + \frac{2}{3} A \wedge A \wedge A)$ 。

By investigating the local expression of Chern-Simons invariant, we can get some other manifestations of anomalies. For this discussion, as G is simply connected we can assign $F = 0$ for flat gauge field. In the M_3 bulk, our gauge field $A(t)$ is flat. On a local patch U_i , we have

$A \wedge A = \frac{k}{2} \text{tr}(A \wedge F - \frac{1}{3} A \wedge A \wedge A)$ on a local patch of M_3 . For simplicity, we restrict to theories whose Chern-Simons invariants used in anomaly computations originate from integral classes in $H^4(BG, \mathbb{Z})$ (cf. Eq. (C11) in Appendix C). This condition constrains the allowed minimal level k , which depends on G . We do not discuss these cases individually and omit the subscript k for brevity [36].

The interpolation from A_Δ to A_Δ is illustrated in the leftmost graph in Figure 2. It is not hard to see that by our convention, the configuration in the front is equivalent to triangle and the one in the back is equivalent to triangle . Since $\mathcal{A}[A_\Delta; \Lambda]$ is clearly independent from small deformations of the mapping cylinder, i.e. the details of g , as long as boundary configurations A_Δ and A_Δ are fixed, we further glue the vertices of triangle with the ones with the same labels in triangle at the two copies $M_2 \times \{0\}$ and $M_2 \times \{1\}$ to arrive at the configuration in the middle of Figure 2. This is the difference between the corresponding webs of defects and the middle configuration can be further deformed to be a decorated tetrahedron as in the rightmost of Figure 2, where the directions of the defects and of the gauge transformations are set by our convention (4). This process can be understood as the anomaly-inflow—very similar to the finite group case [16]—i.e. an extra 3-simplex can be attached in the bulk to cancel the anomaly and the decoration of the tetrahedron in the bulk describes the pullback of the 3-simplex $\langle e, g_1, (g_1 g_2), (g_1 g_2 g_3) \rangle$ in BG^δ to the bulk. Here G^δ is the Lie group G with discrete topology and BG^δ is the *Eilenberg-MacLane* space $K(G, 1)$ classifying flat G -bundles, we denote the classifying map by $\varphi^\delta: M_3 \rightarrow BG^\delta$ [37].

Similar to the finite group case, in terms of the classifying space, the anomaly is obtained by evaluating a cohomology class $[\Omega] \in H^3(BG^\delta, \mathbb{R}/\mathbb{Z})$ on $\langle e, g_1, (g_1 g_2), (g_1 g_2 g_3) \rangle$. Here we treat the coefficient $U(1) = \mathbb{R}/\mathbb{Z}$ as an additive group. Pullback with Ω with φ^δ and integrate over the 3D spacetime, we obtain the Dijkgraaf-Witten phase

$$\mathcal{A}(A_\Delta; \Lambda) = \int_{M_3} (\varphi^\delta)^*(\Omega), \quad (8)$$

in other words, we claim that Ω found by F-move pulls back to the Chern-Simons invariant CS . See the proof of (8) in Appendix C.

$A_i := A|_{U_i} = \tilde{g}_i^{-1} d\tilde{g}_i$ for some $\tilde{g}_i: U_i \rightarrow G$ and $\int_{U_i} -\frac{k}{6} \text{tr}(A_i \wedge A_i \wedge A_i) = \int_{U_i} k \tilde{g}_i^* \omega_3 = \int_{\tilde{g}_i(U_i)} k \omega_3$ (9)

where $\tilde{g}_i^*$ is the pull-back from $\Omega^*(G)$ to $\Omega^*(U_i)$ and

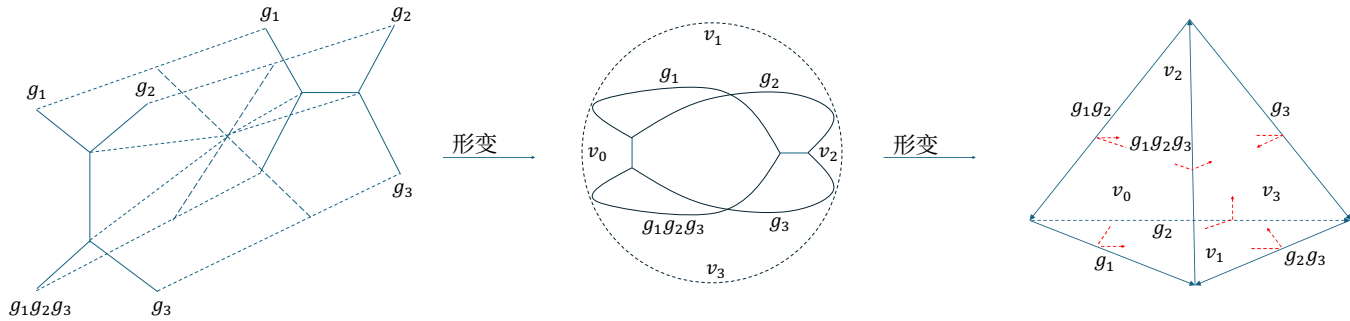


图2. 组合的 $\triangle$ 与 $\triangle$ 构型的形变。图中的群乘法均取自右侧。

$\tilde{g}_i^* \omega_3 = \frac{i^2}{3!} \text{tr}(\tilde{g}_i^{-1} d\tilde{g}_i)^3$, 其中 ω_3 是 G 上的左不变3-形式。对于紧致、连通、单连通且简单的 G, ω_3 , G 生成德拉姆上同调 $H_{\text{dR}}^3(G, \mathbb{R}) \cong \mathbb{R}$, 它也是奇异上同调 $H_{\text{singular}}^3(G, \mathbb{Z}) \cong \mathbb{Z}$ 中的一个整系数类的像[38]。此外, 任意三维流形上的此类 G -丛必然是平凡的。因此, $\tilde{g}_i = \tilde{g}$ 在全局上有定义, 异常可计算为:

$$\mathcal{A}(A_{\times}; \Lambda) = \int_{\tilde{g}(M_3)} k \omega_3. \quad (10)$$

式 (10) 的右端现在有了另一种解释, 即未规范化的二维 Wess-Zumino-Witten 模型的作用。事实上, 这是异常匹配的一个事实[2, 7, 39]。左不变3-形式 ω_3 作为悬挂 (C16) (其逆称为纤维 G 上 CS 沿丛纤维化 $G \rightarrow P \rightarrow M_3$ 的转移 [29, 35]) 的出现, 最初在[40]中讨论; 另见与异常相关的讨论[29, 41-43]。

另一方面, 我们取 $\langle X, Y \rangle = \text{tr}(XY)$ 为李代数 $\mathfrak{g}$ (属于 G) 的 Killing 形式, 则

$$-\frac{k}{6} \text{tr}(A \wedge A \wedge A) = -\frac{k}{12} \langle A, [A, A] \rangle, \quad (11)$$

其中我们认识到, $\langle \cdot, [\cdot, \cdot] \rangle$ 是味代数 $\mathfrak{g}$ 的 3rd 李代数上同调 $H^3(\mathfrak{g}, \mathbb{R})$ 的生成元。我们看到, 群 $H^3(\mathfrak{g}, \mathbb{R})$ 也刻画了平直规范化中的异常局部密度。

由于我们的 G 取为紧致、连通且单的, 我们总有 $H^3(\mathfrak{g}, \mathbb{R}) \cong H_{\text{dR}}^3(G, \mathbb{R}) \cong \mathbb{R}$ 。由于这些3-形式具有整数周期, 它们是 $H_{\text{dR}}^3(G, \mathbb{R})$ 中格 $\mathbb{Z} \cong H_{\text{dR}}^3(G, \mathbb{Z})$ 的像。系数交换映射事实上是单射[44]

$$H_{\text{singular}}^3(G, \mathbb{Z}) \cong H_{\text{dR}}^3(G, \mathbb{Z}) \hookrightarrow H_{\text{dR}}^3(G, \mathbb{R}) \cong H^3(\mathfrak{g}, \mathbb{R}). \quad (12)$$

回顾悬挂映射(C16)

$$\tau : H^4(BG, \mathbb{Z}) \longrightarrow H_{\text{singular}}^3(G, \mathbb{Z}) \quad (13)$$

若 G 还为单连通, 则它也是同构。对于这些群, 我们有

$$H_{\text{singular}}^3(G, \mathbb{Z}) \cong H^4(BG, \mathbb{Z}) \xrightarrow{\kappa^\delta} H^3(BG^\delta, \mathbb{R}/\mathbb{Z}), \quad (14)$$

其中最后的映射 (C9) 也是单射[44]。

额外维度的出现。此外, 还有一个同态 $w : H^3(\mathfrak{g}, \mathbb{R}) \rightarrow H^3(B\bar{G}, \mathbb{R})$, 其中 $B\bar{G}$ 是平坦 G -平凡丛的分类空间[45]。关于 $B\bar{G}$, 一个特别有趣的特征是, 可以将一个平凡化明确写成 $\bar{G} = \{(g, \ell) \in G^\delta \times G^I \mid \ell(0) = g, \ell(1) = e\}$ 的一个元素, 其中 G^δ 是赋予离散拓扑的 G , 并配备典范映射 $\iota : G^\delta \rightarrow G$ 以及 $I := [0, 1]$ 。所对应的 $G^I := \text{Map}(I, G)$

更确切地说, 我们考虑一般二维时空流形 M_2 上的缺陷网络, 该网络通过庞加莱对偶定义了一个平坦的 G -丛 M_2 , 其由分类映射 $f^\delta : M_2 \rightarrow BG^\delta$ 的同伦类确定, 且还有一个由零同伦映射 $f^n : M_2 \rightarrow BG$ 给出的平凡丛。利用 ι 、 $B\iota : BG^\delta \rightarrow BG$ 的提升, 通过映射路径空间构造[46], 映射 $\bar{f} : (x, t) \mapsto (f^\delta(x), \gamma_x(t)) \in BG^\delta \times BG$, 其中 $\gamma_x(0) = B\iota(f^\delta(x))$ 定义了从 f^δ 到 f^n 的同伦。等价地, $\bar{f}$ 是从 $M_2 \times [0, 1]$ 到 $B\bar{G} \subset BG^\delta \times BG$ 的映射。因此, 我们有: —

$$\int_{\bar{f}(M_2 \times I)} w(\omega^3) = \int_{M_2 \times I} \bar{f}^* \circ w(\omega^3) \quad (15)$$

其中 $\bar{f}^*$ 是 $\bar{f}$ 的拉回。 —

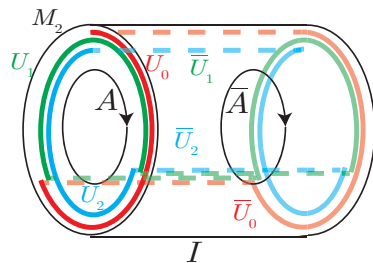


图 3. M_2 的开覆盖 U_k ($k = 0, 1, 2$) 以及 $M_2 \times I$ 的开覆盖 $\bar{U}_k$ ($k = 0, 1, 2$) 的展示。

为在更初等的层面研究抽象映射 $\bar{f}^* \circ w$, 为简单起见, 我们固定 $\pi_1(M_2) = \mathbb{Z}$ (例如令 $M_2 \cong S^1 \times \mathbb{R}$)。我们使用开覆盖 $U_k = \left(\frac{2k\pi}{3}, \frac{(2k+4)\pi}{3}\right)$

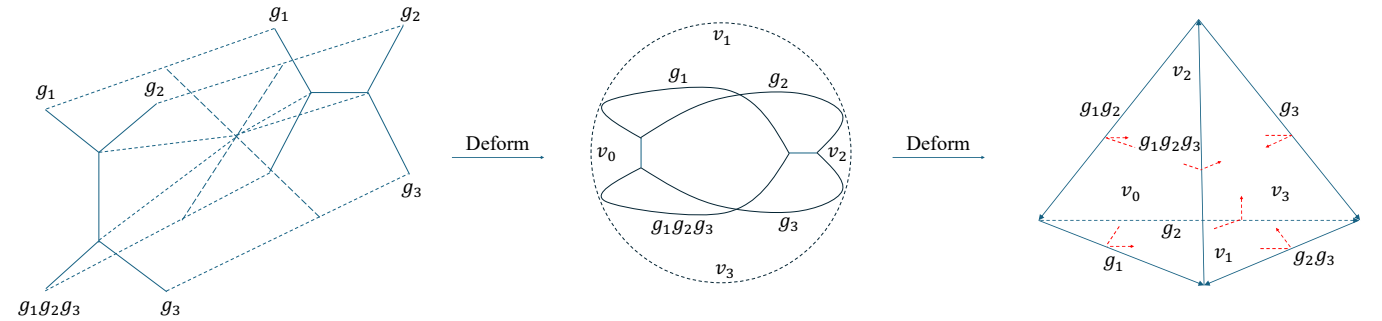


FIG. 2. The deformations of the combined $\triangleleft$ and $\triangleright$ configuration. The group multiplications in this figure are all taken from the right.

$\tilde{g}_i^* \omega_3 = \frac{i^2}{3!} \text{tr}(\tilde{g}_i^{-1} d\tilde{g}_i)^3$ with ω_3 being a *left-invariant 3-form* on G . For a compact, connected, simply connected and simple G , ω_3 generates the de Rham cohomology $H_{\text{dR}}^3(G, \mathbb{R}) \cong \mathbb{R}$, it is also the image of an integral class in the singular cohomology $H_{\text{singular}}^3(G, \mathbb{Z}) \cong \mathbb{Z}$ [38]. Furthermore, such G -bundle on any 3-manifold is necessarily trivial. Hence $\tilde{g}_i = \tilde{g}$ is globally defined, and the anomaly can be calculated as:

$$\mathcal{A}(A_{\times}; \Lambda) = \int_{\tilde{g}(M_3)} k \omega_3. \quad (10)$$

The right-hand side of (10) now has another interpretation, namely, the action of ungauged 2D *Wess-Zumino-Witten* model. Indeed, this is a fact of anomaly matching [2, 7, 39]. The appearance of the left invariant 3-form ω_3 as a *suspension* (C16) (the inverse is referred to as *transgression* [29, 35]) of CS on the fiber G via the bundle fibration $G \rightarrow P \rightarrow M_3$ was discussed originally in [40], see also discussions [29, 41–43] related to anomalies.

On the other hand, we take $\langle X, Y \rangle = \text{tr}(XY)$ to be the Killing form of the Lie algebra $\mathfrak{g}$ of G then

$$-\frac{k}{6} \text{tr}(A \wedge A \wedge A) = -\frac{k}{12} \langle A, [A, A] \rangle, \quad (11)$$

where we recognize that $\langle \cdot, [\cdot, \cdot] \rangle$ is the generator of the 3rd *Lie algebra cohomology* $H^3(\mathfrak{g}, \mathbb{R})$ of the flavor algebra $\mathfrak{g}$. We see that the group $H^3(\mathfrak{g}, \mathbb{R})$ also classifies the local density of anomaly in flat gauging.

Since our G is taken to be compact, connected and simple, we always have $H^3(\mathfrak{g}, \mathbb{R}) \cong H_{\text{dR}}^3(G, \mathbb{R}) \cong \mathbb{R}$. As the 3-forms have integral periods, they are the image of the lattice $\mathbb{Z} \cong H_{\text{dR}}^3(G, \mathbb{Z})$ inside $H_{\text{dR}}^3(G, \mathbb{R})$. The coefficient exchanging map is in fact injective [44]

$$H_{\text{singular}}^3(G, \mathbb{Z}) \cong H_{\text{dR}}^3(G, \mathbb{Z}) \hookrightarrow H_{\text{dR}}^3(G, \mathbb{R}) \cong H^3(\mathfrak{g}, \mathbb{R}). \quad (12)$$

Recall that the suspension map (C16)

$$\tau : H^4(BG, \mathbb{Z}) \longrightarrow H_{\text{singular}}^3(G, \mathbb{Z}) \quad (13)$$

is also an isomorphism if G is in addition simply connected. For these groups, we have

$$H_{\text{singular}}^3(G, \mathbb{Z}) \cong H^4(BG, \mathbb{Z}) \xrightarrow{\kappa^\delta} H^3(BG^\delta, \mathbb{R}/\mathbb{Z}), \quad (14)$$

where the last map (C9) is also an injection [44].

The emergence of extra dimension. Moreover, there is a homomorphism $w : H^3(\mathfrak{g}, \mathbb{R}) \rightarrow H^3(B\bar{G}, \mathbb{R})$ where $B\bar{G}$ is the *classifying space of flat G -trivial bundles* [45]. One particularly interesting feature about $B\bar{G}$ is that a trivialization can be explicitly written down as an element of $\bar{G} = \{(g, \ell) \in G^\delta \times G^I \mid \ell(0) = g, \ell(1) = e\}$ where G^δ is G equipped with discrete topology together with the canonical map $\iota : G^\delta \rightarrow G$ and $G^I := \text{Map}(I, G)$ for $I := [0, 1]$.

More precisely, we consider a web of defects on a general 2D spacetime manifold M_2 which, by Poincaré duality, defines a flat G -bundle over M_2 determined up to isomorphism by the homotopy class of the *classifying map* $f^\delta : M_2 \rightarrow BG^\delta$ and a trivial bundle given by a null-homotopic map $f^n : M_2 \rightarrow BG$. Using the lift of ι , $B\iota : BG^\delta \rightarrow BG$, the map $\bar{f} : (x, t) \mapsto (f^\delta(x), \gamma_x(t)) \in BG^\delta \times BG$ where $\gamma_x(0) = B\iota(f^\delta(x))$ defines a homotopy from f^δ to f^n by the *mapping path space construction* [46]. Equivalently, $\bar{f}$ is a map from $M_2 \times [0, 1]$ to $B\bar{G} \subset BG^\delta \times BG$. Therefore, we have:

$$\int_{\bar{f}(M_2 \times I)} w(\omega^3) = \int_{M_2 \times I} \bar{f}^* \circ w(\omega^3) \quad (15)$$

where $\bar{f}^*$ is the pull-back of $\bar{f}$.

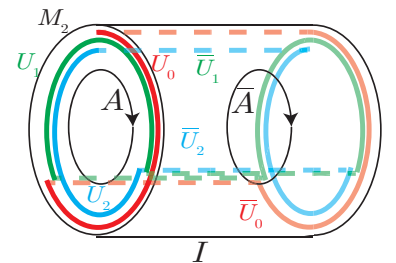


FIG. 3. The showing of open cover U_k ($k = 0, 1, 2$) of M_2 as well as the open cover $\bar{U}_k$ ($k = 0, 1, 2$) of $M_2 \times I$.

To investigate the abstract map $\bar{f}^* \circ w$ at a more elementary level, we fix $\pi_1(M_2) = \mathbb{Z}$ for simplicity (e.g. let $M_2 \cong S^1 \times \mathbb{R}$). We use the open cover $U_k = \left(\frac{2k\pi}{3}, \frac{(2k+4)\pi}{3}\right)$

(模 2π) 的 $S^1, k = 0, 1, 2$, 并在平凡丛上定义平坦联络 $A = \frac{i\alpha}{2\pi} d\theta$ ($\alpha \in i\mathfrak{g}$), 其在从 U_i 到 U_j 的 $U_{ij} := U_i \cap U_j$ 上的过渡函数为 $g_{ij} \equiv 1$ 。我们在图 3 中绘制时空几何和斑块。在每个由 $\theta_k \in (0, \frac{4}{3}\pi)$, A_k 参数化的 U_k 上进行规范变换 $u_k = e^{-\frac{i\alpha}{2\pi}(\theta_k + \frac{2k\pi}{3})}$ 后, $u_k^{-1}A_k u_k + u_k^{-1}du_k = 0$ 使其消失。同时, $U_{i,i+1}$ 上的过渡函数变为 $g'_{i,i+1} = u_i^{-1}g_{i,i+1}u_{i+1} = u_i^{-1}u_{i+1}$ 。由于对于 i 模 3, $\theta_i(p) = \theta_{i+1}(p) + \frac{2\pi}{3}, \forall p \in U_{i,i+1}$, 我们有:

$$g'_{01}(U_{01}) = g'_{12}(U_{12}) = 1, g'_{20}(U_{20}) = e^{i\alpha}. \quad (16)$$

因此, 尽管 u_k 使 A 平凡化, 它却产生了由 α 参数化的非平凡过渡函数。为了将 A 提升到 $M_2 \times I$, 我们定义 $\bar{u}_k = e^{-\frac{i\alpha}{2\pi}(\theta_k + \frac{2k\pi}{3})f(t)}$, 并令 $f(0) = 1$ 和 $f(1) = 0$ 对于 $t \in I$ 是从 $\bar{U}_k := U_k \times \mathbb{R} \times I$ 到 G 的函数。我们进一步考虑 $\bar{U}_k$ 上的规范场 $\bar{A}_k = \bar{u}_k^{-1}A_k\bar{u}_k + \bar{u}_k^{-1}d\bar{u}_k$ 以及 $\bar{U}_{ij}$ 上的过渡函数 $\bar{g}_{ij} = \bar{u}_i^{-1}\bar{u}_j$ 。定义在 $M_2 \times I$ (的一个开覆盖) 上的对 $(\bar{g}_{ij}, \bar{A}_i)$, 将 $M_2 \times \{0\}$ 上具有非平凡过渡函数 (16) 的丛平凡化为 $M_2 \times \{1\}$ 上带有非零平坦联络的平凡丛。在此例中, $\bar{g}_{20} = e^{i\alpha f(t)}$ 为 $\bar{G}$ 的一个元素 ($e^{i\alpha}, \ell$) 提供了具体的物理实现。

由于带有 $\bar{g}_k = u_k^{-1}\bar{u}_k$ 的 $\bar{A}_k = \bar{g}_k^{-1}d\bar{g}_k$ 是平坦的, 我们可以将 (9) 中的 (A, U) 替换为 $(\bar{A}_k, \bar{U}_k)$ 。再结合限制在 $\bar{U}_k$ 上的 (15) 以及适当的归一化选择, 我们得到 47:

$$\bar{f}^* \circ w(\omega_3)|_{\bar{U}_k} = \bar{g}^*(\omega_3)|_{\bar{U}_k} = CS[\bar{A}]|_{\bar{U}_k}. \quad (17)$$

在物理上, 这意味着, 't Hooft 反常的数据可以由 $M_2 \times I$ 上一个平坦规范场的陈-西蒙斯作用量刻画, 该规范场在 $M_2 \times \{0\}$ 处具有非平凡过渡函数, 并在 $M_2 \times \{1\}$ 处实现平凡化, 但代价是产生非零平坦联络。此外, 由 $H^3(\mathfrak{g}, \mathbb{R})$ 对 't Hooft 反常的刻画自然要求反常理论出现额外维度, 由于 $\bar{f}^* \circ w$ 的结构, 这表现为 I 。

$U(1)$ 味对称性及其 't Hooft 反常。数学上谨慎的读者或许已经注意到, 之前导出 $H^3(G, \mathbb{R})$ 的计算对于 $U(1)$ 并不成立, 因为 $H^3(U(1), \mathbb{R})$ 和 $H^3(\mathfrak{u}(1), \mathbb{R})$ 都为零。此外, 产生 $H^4(BU(1), \mathbb{Z}) \cong \mathbb{Z}$ 的 $U(1)$ 的反常多项式为 $CS[A] = A \wedge dA$, 其中三次项 $\frac{1}{3}A \wedge A \wedge A$ 缺失。为得到非零反常, 我们指出, 在 (5) 中的结点 $(x, y) = (0, 0)$ 处场强不为零。

对于 $G = U(1)$ 的情形, 有 $f(\alpha, \beta) = (\alpha + \beta) \bmod 2\pi\mathbb{Z} \equiv [\alpha + \beta]$ 。与三价结点对偶的平坦规范场为

$$A = i\alpha\delta(x)H(y)dx + i(\beta H(-x) + [\alpha + \beta]H(x))\delta(y)dy, \quad (18)$$

相应的场强为

$$F = dA = i([\alpha + \beta] - \alpha - \beta)\delta(x)\delta(y)dx \wedge dy \quad (19)$$

在结点处非零。现在, 为计算 $U(1)$ 味对称性的 't Hooft 反常, 我们将之前的计算应用于图4中的构型

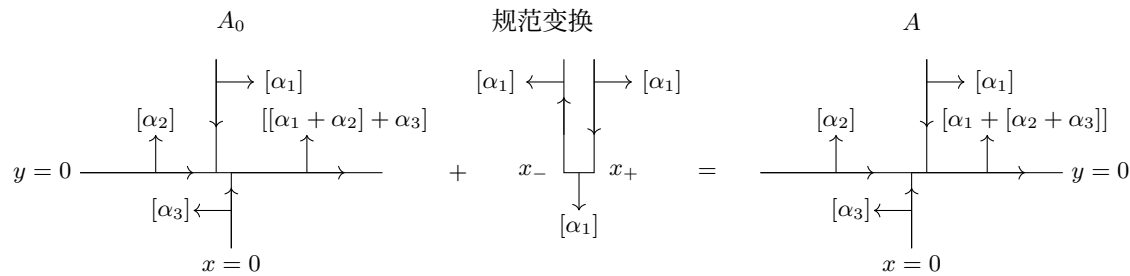


图4. 具有 $U(1)$ 味对称性的理论中缺陷线的F变换。携带 $e^{i\alpha} \in U(1)$ 的拓扑线标记为 $[\alpha] := \alpha \bmod 2\pi\mathbb{Z}$, 箭头表示规范变换的方向, 与 (5) 中的方向一致。

容易看出, 图4中最左侧和最右侧的构型分别在拓扑上等价于A和A。因此, 我们的任务是求出它们的庞加莱对偶 A_Δ 和 A_Δ , 以及规范变换参数 $d\epsilon \equiv id\Lambda = A_\Delta - A_\Delta$ 。实际上, 利用图4中间的构型并假设

$$\epsilon = -i\alpha_1(H(x_-) - H(x_+))H(y) \quad (20)$$

(modulo 2π) of $S^1, k = 0, 1, 2$, and define the flat connection $A = \frac{i\alpha}{2\pi} d\theta$ ($\alpha \in i\mathfrak{g}$) on a trivial bundle with the transition function $g_{ij} \equiv 1$ on $U_{ij} := U_i \cap U_j$ from U_i to U_j . We plot the spacetime geometry and patches in Figure 3. Under gauge transformation $u_k = e^{-\frac{i\alpha}{2\pi}(\theta_k + \frac{2k\pi}{3})}$ on each U_k parametrized by $\theta_k \in (0, \frac{4}{3}\pi)$, A_k vanishes because $u_k^{-1}A_k u_k + u_k^{-1}du_k = 0$. Meanwhile, the transition function on $U_{i,i+1}$ becomes $g'_{i,i+1} = u_i^{-1}g_{i,i+1}u_{i+1} = u_i^{-1}u_{i+1}$. Since $\theta_i(p) = \theta_{i+1}(p) + \frac{2\pi}{3}, \forall p \in U_{i,i+1}$ for $i \bmod 3$, we have:

$$g'_{01}(U_{01}) = g'_{12}(U_{12}) = 1, g'_{20}(U_{20}) = e^{i\alpha}. \quad (16)$$

Hence, while u_k trivializes A , it yields non-trivial transition functions parametrized by α . To uplift A to $M_2 \times I$, we define $\bar{u}_k = e^{-\frac{i\alpha}{2\pi}(\theta_k + \frac{2k\pi}{3})f(t)}$ with $f(0) = 1$ and $f(1) = 0$ for $t \in I$ as a function from $\bar{U}_k := U_k \times \mathbb{R} \times I$ to G . We further consider the gauge field $\bar{A}_k = \bar{u}_k^{-1}A_k\bar{u}_k + \bar{u}_k^{-1}d\bar{u}_k$ on $\bar{U}_k$ and the transition function $\bar{g}_{ij} = \bar{u}_i^{-1}\bar{u}_j$ on $\bar{U}_{ij}$. The pair $(\bar{g}_{ij}, \bar{A}_i)$, defined on (an open cover of) $M_2 \times I$, leads to a trivialization of a bundle with non-trivial transition function (16) over $M_2 \times \{0\}$ to a trivial bundle with non-zero flat connection over $M_2 \times \{1\}$. In this example, $\bar{g}_{20} = e^{i\alpha f(t)}$ provides a concrete physical realization of an element $(e^{i\alpha}, \ell)$ of $\bar{G}$.

Since $\bar{A}_k = \bar{g}_k^{-1}d\bar{g}_k$ with $\bar{g}_k = u_k^{-1}\bar{u}_k$ is flat, we can replace (A, U) in (9) by $(\bar{A}_k, \bar{U}_k)$. Together with (15) restricted to $\bar{U}_k$ and a suitable choice of normalization, we are led to [47]:

$$\bar{f}^* \circ w(\omega_3)|_{\bar{U}_k} = \bar{g}^*(\omega_3)|_{\bar{U}_k} = CS[\bar{A}]|_{\bar{U}_k}. \quad (17)$$

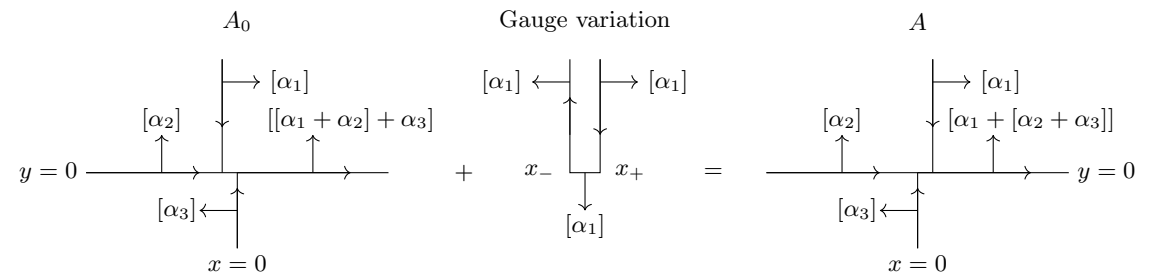


FIG. 4. The F-move of defect lines in a theory with $U(1)$ flavor symmetry. A topological line carrying $e^{i\alpha} \in U(1)$ is labeled by $[\alpha] := \alpha \bmod 2\pi\mathbb{Z}$, with an arrow labeling the direction of gauge transformation in the same sense as in (5).

It is easy to see that the left-most and the right-most configurations in Figure 4 are topologically equivalent to $\nwarrow$ and $\nearrow$, respectively. Hence our task is to find their Poincaré duals $A_{\nwarrow}$ and $A_{\nearrow}$, and the gauge transformation parameter $d\epsilon \equiv id\Lambda = A_{\nwarrow} - A_{\nearrow}$. Actually, using the configuration in the middle of Figure 4 and assum-

Physically, this means that the data of 't Hooft anomaly can be captured by the Chern-Simons action of a flat gauge field on $M_2 \times I$ with non-trivial transition functions at $M_2 \times \{0\}$ which trivializes at $M_2 \times \{1\}$ at the price of yielding non-zero flat connection. Moreover, the characterization of 't Hooft anomaly by $H^3(\mathfrak{g}, \mathbb{R})$ naturally requires the emergence of an extra dimension of the anomaly theory, manifests itself as I , because of the structure of $\bar{f}^* \circ w$.

$U(1)$ flavor symmetry and its 't Hooft anomaly. A mathematically cautious reader might have already noticed that the previous calculation leading to $H^3(G, \mathbb{R})$ fails for $U(1)$, since both $H^3(U(1), \mathbb{R})$ and $H^3(\mathfrak{u}(1), \mathbb{R})$ vanish. Moreover, the anomaly polynomial for $U(1)$ that generates $H^4(BU(1), \mathbb{Z}) \cong \mathbb{Z}$ is $CS[A] = A \wedge dA$, where the cubic term $\frac{1}{3}A \wedge A \wedge A$ is absent. To get a non-zero anomaly, we comment that there is a non-zero field strength at the junction point $(x, y) = (0, 0)$ in (5).

For the case of $G = U(1)$, we have $f(\alpha, \beta) = (\alpha + \beta) \bmod 2\pi\mathbb{Z} \equiv [\alpha + \beta]$. The flat gauge field dual to the trivalent junction is then

$$A = i\alpha\delta(x)H(y)dx + i(\beta H(-x) + [\alpha + \beta]H(x))\delta(y)dy, \quad (18)$$

and the corresponding field strength

$$F = dA = i([\alpha + \beta] - \alpha - \beta)\delta(x)\delta(y)dx \wedge dy \quad (19)$$

is non-vanishing at the junction point. Now to compute the 't Hooft anomaly of $U(1)$ flavor symmetry, we apply the previous computation to the configuration in Figure 4.

ing $\alpha < 2\pi$ hence $[\alpha_1] = \alpha_1$, one can show that (see the details in Appendix A):

$$\epsilon = -i\alpha_1(H(x_-) - H(x_+))H(y) \quad (20)$$

由此可将 ’t Hooft 反常计算为

$$\mathcal{A}[A_{\triangleleft}; \epsilon] = \frac{1}{(2\pi)^2} \int_{\mathbb{R}^2} \epsilon dA_{\triangleleft} = \frac{1}{(2\pi)^2} \alpha_1 (\alpha_2 + \alpha_3 - [\alpha_2 + \alpha_3]). \quad (21)$$

将

$$f(e^{i\alpha_1}, e^{i\alpha_2}, e^{i\alpha_3}) := e^{2\pi i A[A_{\triangleleft}; \epsilon]} = e^{i\alpha_1 (\alpha_2 + \alpha_3 - [\alpha_2 + \alpha_3]) / (2\pi)} \quad (22)$$

视为映射 $f: G^3 \rightarrow U(1)$, 可以验证 $\delta f = 0$, 其中 δ 是 $U(1)^\delta$ 的李群上同调的余边界算子（附录A中给出了具体计算）。

映射 22 显然不是连续的, 这与使用离散拓扑的群上同调 $H^3(U(1)^\delta, U(1))$ 相一致。将其与图 1 中的相位因子比较可见, (22) 恰好是 $H^3(\mathbb{Z}_m, U(1))$ 的生成元的 $m \rightarrow \infty$ 极限。进一步可以证明, 该映射定义了一个非平凡的代数 3-上循环 [48]。

在群层面, $\mathbb{Z}_m$ 的直接极限是 $\mathbb{Q}/\mathbb{Z}$, 即所有单位根的并, 作为 $U(1)^\delta$ 的挠子群。众所周知, $H^3(\mathbb{Q}/\mathbb{Z}, U(1)) = \widehat{\mathbb{Z}}$, 其中 $\widehat{\mathbb{Z}}$ 是拟有限整数群, 且是 $\mathbb{Z}_m$ 的逆极限（例如见 [49]）。实际上, 一个元素 $\widehat{k} \in \widehat{\mathbb{Z}}$ 可以描述为其对所有整数 N 的取值（即在 $\mathbb{Z}_N$ 中的像）模 N 的集合。令 $x = \frac{p}{q}, y = \frac{r}{s}, z = \frac{t}{u}$ 为 $\mathbb{Q}/\mathbb{Z}$ 中满足 $\gcd(p, q) = \gcd(r, s) = \gcd(t, u) = 1$ 和 $0 \leq p < q, 0 \leq r < s$ 的元素, $0 \leq t < u$ 满足 $p, r, t \in \mathbb{Z}$ 和 $q, s, u \in \mathbb{Z} \setminus \{0\}$ 。我们以 M 表示 q, s 和 u 的最小公倍数。由 $\widehat{k}$ 表示的 $H^3(\mathbb{Q}/\mathbb{Z}, U(1))$ 中的 3-上循环可明确写为相位

$$\omega_{\widehat{k}}(x, y, z) = \exp \left(2\pi i \frac{(\widehat{k} \bmod M) p}{q} \lambda(r, s, t, u) \right), \quad (23)$$

其中, 当 $\frac{r}{s} + \frac{t}{u} > 1$ 时 $\lambda(r, s, t, u) = 1$, 否则为 $\lambda(r, s, t, u) = 0$ 。

由表达式 23 可见, $U(1)^\delta$ 的 $\mathbb{Q}/\mathbb{Z}$ 部分及其为 $\widehat{\mathbb{Z}}$ 的三阶上同调群刻画了所有 $\mathbb{Z}_m \subset U(1)^\delta$ 的反常。上述群论讨论及 $H^3(U(1)^\delta, U(1)) \supset H^4(BU(1), \mathbb{Z})$ 的计算详见附录 D

高维推广。我们已明确推导出拓扑缺陷的 F-移动与连续对称群 G 的平坦规范化异常多项式之间的联系, 针对的是 $d = 2$ 情形。注意, 如果取图2右图的对偶图, 我们得到一个 3-单纯形 $v_0 v_1 v_2 v_3$, 其顶点为 v_i ($i = 0, \dots, 3$), 而边对应于 G 的元素。该图景可轻易推广到 $d = 2k$ 时空维度, 此时对偶 3-单纯形 $v_0 v_1 v_2 v_3$ 被对偶 $(2k+1)$ -单纯形 $v_0 v_1 \dots v_{2k}$ 所取代。对 $v_0 v_1 \dots v_{2k}$ 取对偶, 便得到 F-移动的高维推广, 如 Dijkgraaf-Witten 理论 [29] 中所示。

李代数上同调 $H^3(\mathfrak{g}, \mathbb{R})$ 和群上同调 $H^3(U(1)^\delta, U(1))$ 应可直接推广为 $H^{2k+1}(\mathfrak{g}, \mathbb{R})$ 和 $H^{2k+1}(U(1)^\delta, U(1))$ 。我们在附录 B 中给出了半单李代数 $\mathfrak{g}, H^{2k+1}(U(1)^\delta, U(1))$ 对应一组 $U(1)^0$ -形式对称群时 $H^{2k+1}(\mathfrak{g}, \mathbb{R})$ 的显式生成元, 以及它们与异常多项式的对应关系。

致谢

我们感谢陈进、Tachikawa Yuji、万哲彦、王鑫、王兆龙和郑云琴的有益讨论。YZ 感谢 Mikhail Kapranov 对离散化李群上同调的有益解释。JT 和 YNW 感谢彭桓武中心基础理论研究中心和西北大学主办 6th 全国场论与弦理论研讨会, 本项目取得的许多进展均是在会议期间完成的。RL 和 YNW 获得中国国家自然科学基金委项目（批准号: 12175004、12422503）以及中国科学技术协会青年人才托举工程项目（2024QNRC001）的资助。YNW 还获得中国国家自然科学基金委项目（批准号: 12247103）的资助。QJ 获得韩国国家研究基金会（NRF）项目（批准号: RS-2024-00405629）和韩国科学技术院 Jang Young-Sil Fellow Program 的资助。YZ 获得日本文部科学省 WPI 计划在东京大学 Kavli IPMU 的资助, 以及中国国家自然科学基金委（NSFC）项目（批准号: 12305077）的资助。

This enables us to compute the ’t Hooft anomaly to be

$$\mathcal{A}[A_{\triangleleft}; \epsilon] = \frac{1}{(2\pi)^2} \int_{\mathbb{R}^2} \epsilon dA_{\triangleleft} = \frac{1}{(2\pi)^2} \alpha_1 (\alpha_2 + \alpha_3 - [\alpha_2 + \alpha_3]). \quad (21)$$

Viewing

$$f(e^{i\alpha_1}, e^{i\alpha_2}, e^{i\alpha_3}) := e^{2\pi i A[A_{\triangleleft}; \epsilon]} = e^{i\alpha_1 (\alpha_2 + \alpha_3 - [\alpha_2 + \alpha_3]) / (2\pi)} \quad (22)$$

as a map $f: G^3 \rightarrow U(1)$, one can check that $\delta f = 0$ where δ is the coboundary operator for the Lie group cohomology of $U(1)^\delta$ (see an explicit computation in the appendix A).

The map (22) is obviously not continuous, which aligns with the usage of group cohomology $H^3(U(1)^\delta, U(1))$ with discrete topology. Comparing to the phase factor in Figure 1, we see that (22) is exactly the $m \rightarrow \infty$ limit of the generator of $H^3(\mathbb{Z}_m, U(1))$. One can further show that this map defines a non-trivial algebraic 3-cocycle [48].

At the group level, the direct limit of $\mathbb{Z}_m$ is $\mathbb{Q}/\mathbb{Z}$, the union of all roots of unity, as the torsion subgroup of $U(1)^\delta$. It is well-known that $H^3(\mathbb{Q}/\mathbb{Z}, U(1)) = \widehat{\mathbb{Z}}$, where $\widehat{\mathbb{Z}}$ is the group of profinite integers, and it is the inverse limit of $\mathbb{Z}_m$ (see for instance [49]). In practice, an element $\widehat{k} \in \widehat{\mathbb{Z}}$ can be described as a collection of its values (its images in $\mathbb{Z}_N$) modulo N for all integer N . Let $x = \frac{p}{q}, y = \frac{r}{s}, z = \frac{t}{u}$ be elements in $\mathbb{Q}/\mathbb{Z}$ such that $\gcd(p, q) = \gcd(r, s) = \gcd(t, u) = 1$ and $0 \leq p < q, 0 \leq r < s, 0 \leq t < u$ with $p, r, t \in \mathbb{Z}$ and $q, s, u \in \mathbb{Z} \setminus \{0\}$. We denote by M the least common multiple of q, s and u . The 3-cocycle in $H^3(\mathbb{Q}/\mathbb{Z}, U(1))$ represented by $\widehat{k}$ can be explicitly written as the phase

$$\omega_{\widehat{k}}(x, y, z) = \exp \left(2\pi i \frac{(\widehat{k} \bmod M) p}{q} \lambda(r, s, t, u) \right), \quad (23)$$

where $\lambda(r, s, t, u) = 1$ if $\frac{r}{s} + \frac{t}{u} > 1$ and $\lambda(r, s, t, u) = 0$ otherwise.

From the expression (23), we see that the $\mathbb{Q}/\mathbb{Z}$ part of $U(1)^\delta$ with its third cohomology group being $\widehat{\mathbb{Z}}$ captures the anomalies for all $\mathbb{Z}_m \subset U(1)^\delta$. For the details of the above group theoretical discussions and the computation of $H^3(U(1)^\delta, U(1)) \supset H^4(BU(1), \mathbb{Z})$, see Appendix D.

Higher dimensional generalizations. We have explicitly derived the connection between the F-move of topological defects and the anomaly polynomial for the flat gauging of the continuous symmetry group G , for the $d = 2$ case. Note that if we take the dual graph of the right picture in Figure 2, we obtain a 3-simplex $v_0 v_1 v_2 v_3$ whose vertices are v_i ($i = 0, \dots, 3$) while the edges correspond to elements of G . Such picture can be easily generalized to $d = 2k$ spacetime dimensions, where the dual 3-simplex $v_0 v_1 v_2 v_3$ is replaced by a dual $(2k+1)$ -simplex $v_0 v_1 \dots v_{2k}$. Taking the dual of $v_0 v_1 \dots v_{2k}$, one obtains the higher-dimensional generalization of F-move, as in Dijkgraaf-Witten theory [29].

The Lie algebra cohomology $H^3(\mathfrak{g}, \mathbb{R})$ and group cohomology $H^3(U(1)^\delta, U(1))$ should be straightforwardly generalized to $H^{2k+1}(\mathfrak{g}, \mathbb{R})$ and $H^{2k+1}(U(1)^\delta, U(1))$. We present the explicit generators for $H^{2k+1}(\mathfrak{g}, \mathbb{R})$ for semisimple Lie algebra $\mathfrak{g}$, $H^{2k+1}((U(1)^I)^\delta, U(1))$ for a product of $U(1)$ 0-form symmetry groups, and their correspondence with anomaly polynomials, in Appendix B.

ACKNOWLEDGMENTS

We would like to thank Jin Chen, Yuji Tachikawa, Zheyuan Wan, Xin Wang, Zhaolong Wang and Yunqin Zheng for useful discussions. YZ would like to thanks Mikhail Kapranov for helpful explanations about the cohomology of Lie groups made discrete. JT and YNW would like to thank Peng Huanwu Center for Fundamental Theory and Northwest University for hosting the 6th National Workshop on Fields and Strings where many progress of this project has been made. RL and YNW are supported by National Natural Science Foundation of China under Grant No. 12175004, No. 12422503 and by Young Elite Scientists Sponsorship Program by CAST (2024QNRC001). YNW is also supported by National Natural Science Foundation of China under Grant No. 12247103. QJ is supported by National Research Foundation of Korea (NRF) Grant No. RS-2024-00405629 and Jang Young-Sil Fellow Program at the Korea Advanced Institute of Science and Technology. YZ is supported by WPI Initiative, MEXT, Japan at Kavli IPMU, the University of Tokyo and by National Science Foundation of China (NSFC) under Grant No. 12305077.

[1] G. ’t Hooft, 自然性、手征对称性与自发手征对称性破缺, NATO Sci. Ser. B 59, 135 (1980)

[2] E. Witten, 流代数的整体方面, Nucl. Phys. B 223, 422 (1983)

[3] J. Terning, QCD 的 ’t Hooft 异常匹配, Phys. Rev. Lett. 80, 2517 (1998), arXiv:hep-th/9706074

[4] R. Stora, 连续统规范理论, Conf. Proc. C 7607121, 201 (1976)。

[5] R. Stora, 异常的代数结构与拓朴起源, 载于《规范场论进展》, G. ’t Hooft 等编 (Plenum Press, 1984), 第 543–562 页; 课程讲义发表于 1983 年 9 月 1–15 日法国卡尔斯斯夏季学院。

[6] B. Zumino、Y.-S. Wu 和 A. Zee, 手征异常、高维与微分几何, Nucl. Phys. B 239, 477 (1984)

[7] J. Wess 和 B. Zumino, 异常

[1] G. ’t Hooft, Naturalness, chiral symmetry, and spontaneous chiral symmetry breaking, NATO Sci. Ser. B **59**, 135 (1980).

[2] E. Witten, Global Aspects of Current Algebra, Nucl. Phys. B **223**, 422 (1983).

[3] J. Terning, ’t Hooft anomaly matching for QCD, Phys. Rev. Lett. **80**, 2517 (1998), arXiv:hep-th/9706074.

[4] R. Stora, Continuum Gauge Theories, Conf. Proc. C **7607121**, 201 (1976).

[5] R. Stora, Algebraic structure and topological origin of anomalies, in *Progress in Gauge Field Theory*, edited by G. ’t Hooft *et al.* (Plenum Press, 1984) pp. 543–562, lectures given at Cargèse Summer Inst., Cargèse, France, Sep 1–15, 1983.

[6] B. Zumino, Y.-S. Wu, and A. Zee, Chiral Anomalies, Higher Dimensions, and Differential Geometry, Nucl. Phys. B **239**, 477 (1984).

[7] J. Wess and B. Zumino, Consequences of anomalous

- Ward 恒等式的后果, Phys. Lett. B 37, 95 (1971)
- [8] C. G. Callan, Jr. 和 J. A. Harvey, 弦与畴壁上的异常和费米子零模, Nucl. Phys. B 250, 427 (1985)
- [9] M. F. Atiyah, V. K. Patodi 和 I. M. Singer, 谱不对称性与黎曼几何 1, Math. Proc. Cambridge Phil. Soc. 77, 43 (1975)
- [10] E. Witten, 一个 SU(2) 异常, Phys. Lett. B 117, 324 (1982)
- [11] E. Witten, 整体引力异常, Commun. Math. Phys. 100, 197 (1985)
- [12] E. Witten 和 K. Yonekura, 异常流入与 η 不变量, 载于《Shoucheng Zhang 纪念研讨会》(2019), arXiv:1909.08775 [hep-th]
- [13] X.-G. Wen, 通过对称性保护的平凡序分类规范异常, 并通过拓扑序分类引力异常, Phys. Rev. D 88, 045013 (2013)] arXiv:1303.1803 [hep-th]
- [14] A. Kapustin 和 R. Thorngren, 《规范理论的高阶对称性与有能隙相》, Prog. Math. 324, 177 (2017), arXiv:1309.4721 [hep-th]
- [15] D. S. Freed, 《反常与可逆场论》, Proc. Symp. Pure Math. 88, 25 (2014), arXiv:1404.7224 hep-th
- [16] Y. Tachikawa, 《关于有限子群的规范化》, SciPost Phys. 8, 015 (2020), arXiv:1712.09542 [hep-th]
- [17] L. Bhardwaj 和 Y. Tachikawa, 《关于二维中的有限对称性及其规范化》, JHEP 03, 189 arXiv:1704.02330 [hep-th]
- [18] D. Gaiotto, A. Kapustin, N. Seiberg 和 B. Willett, 《广义全局对称性》, JHEP 02, 172 arXiv:1412.5148 [hep-th]
- [19] J. McGreevy, 《凝聚态物质中的广义对称性》, Ann. Rev. Condensed Matter Phys. 14, 57 (2023) arXiv:2204.03045 [cond-mat.str-el]
- [20] S. Schafer-Nameki, 《关于（非）可逆广义对称性的 ICTP 讲义》, Phys. Rept. 1063, 1 (2024) arXiv:2305.18296 [hep-th]
- [21] L. Bhardwaj, L. E. Bottini, L. Fraser-Taliente, L. Glad-den, D. S. W. Gould, A. Platschorre 和 H. Tillim, 《广义对称性讲义》, Phys. Rept. 1051, 1 (2024) arXiv:2307.07547 [hep-th]
- [22] R. Luo, Q.-R. Wang 和 Y.-N. Wang, 《广义对称性及其应用讲义》, Phys. Rept. 1065, 1 (2024) arXiv:2307.09215 [hep-th]
- [23] S.-H. Shao, 《木已成舟: 不可逆不对称性的 TASI 讲义》, arXiv (2023), arXiv:2308.00747 [hep-th]
- [24] X. Chen, Z.-C. Gu, Z.-X. Liu 和 X.-G. Wen, 《对称性保护拓扑序及其对称群的群上同调》, Phys. Rev. B 87, 155114 (2013) arXiv:1106.4772 [cond-mat.str-el]
- [25] A. Kapustin 和 R. Thorngren, 《三维中离散对称性的反常与群上同调》, Phys. Rev. Lett. 112, 231602 (2014). arXiv:1403.0617 [hep-th]
- [26] A. Kapustin 和 R. Thorngren, 《各种维度中离散对称性的反常与群上同调》, arXiv (2014), arXiv:1404.3230 [hep-th]
- [27] D. V. Else 和 C. Nayak, 《通过对称性在边界上的反常作用对对称性保护拓扑相进行分类》, Phys. Rev. B 90, 235137 (2014) arXiv:1409.5436 [cond-mat.str-el]
- [28] G. W. Moore 和 N. Seiberg, 《经典与量子共形场论》, Commun. Math. Phys. 123, 177 (1989)
- [29] R. Dijkgraaf 和 E. Witten, 《拓扑规范理论与群上同调》, Commun. Math. Phys. 129, 393 (1990)
- [30] C.-M. Chang, Y.-H. Lin, S.-H. Shao, Y. Wang 和 X. Yin, 《二维中的拓扑缺陷线与重整化群流》, JHEP 01, 026 arXiv:1802.04445 [hep-th]
- [31] J. Milnor, 《关于离散化李群的同调》, Commentarii Mathematicae Helvetici 58, 72 (1983)
- [32] [48] 使用 Else 和 Nayak 的方法 27, 为玻色二维理论推导出了相同的 3-上循环表达式。在这一框架下关于费米情形中 $U(1)$ 异常的最新讨论见 50。
- [33] Q. Jia, R. Luo, J. Tian, Y.-N. Wang 和 Y. Zhang, 《范畴连续对称性》, arXiv (2025), arXiv:2509.13170 [hep-th]
- [34] D. S. Freed, M. J. Hopkins, J. Lurie 和 C. Teleman, 《紧李群产生的拓扑量子场论》, 载于《数学中 Raoul Bott 遗产庆典》(2009), arXiv:0905.0731 [math.AT]
- [35] A. Borel, 《李群的拓扑与特征类》, Bulletin of the American Mathematical Society 61, 397 (1955)
- [36] 对于某些紧李群, 例如 $U(1)$ 和 $SO(3)$, 在级别 $k = 1$ 定义 Chern-Simons 不变量需要自旋结构 [29, 51]。在这些情形下, 将 Chern-Simons 项替换为相应的 η -不变量 (或等价地, 规范场配对的二次精化)。这对应于 $H^4(BG, \mathbb{Z})$ 生成元的一半, 因此不能直接作为 $H^4(BG, \mathbb{Z})$ 中的一个类得到。在本文中, 我们只要求时空流形及其延拓具有可定向性。
- [37] 关于 BG^δ 的更多细节见附录 C。关于有限 G 的 BG 的单纯复形构造, 读者可参阅读讲义第 9.1 节: <https://member.ipmu.jp/yuji.tachikawa/lectures/2024-mathphys/notes.pdf>
- [38] 我们采用约定 $\omega_{2n-1} := i^n \frac{(n-1)!}{(2n-1)!} \text{tr}(\tilde{g}^{-1} d\tilde{g})^{2n-1}$, 使其在 $\pi_{2n-1}(G)$ 的生成元上的积分归一化为 1; 对于 $n = 2$, $\pi_3(G)$ 的生成元被映射到 $H_3(G, \mathbb{Z})$ 的生成元 [43]
- [39] S. P. Novikov, 《哈密顿形式与 Morse 理论的多值类比》, Usp. Mat. Nauk 37N5, 3 (1982)
- [40] S.-S. Chern 和 J. Simons, 《特征形式与几何不变量》, Annals Math. 99, 48 (1974)
- [41] E. Witten, 《WZW 与余集模型的全纯因子化》, Commun. Math. Phys. 144, 189 (1992)
- [42] D. S. Freed, 《 π 介子与广义上同调》, J. Diff. Geom. 80, 45 (2008), arXiv:hep-th/0607134
- [43] Y. Lee, K. Ohmori 和 Y. Tachikawa, 《重新审视 Wess-Zumino-Witten 项》, SciPost Phys. 10, 061 (2021), arXiv:2009.00033 [hep-th]
- [44] J. C. Baez 和 A. D. Lauda, 高维代数 V: 2-群, arXiv (2003), arXiv:math/0307200
- [45] S. Morita 和 A. M. Society, 《特征类的几何》。现代数学岩波丛书 (Ameri-can Mathematical Society, 2001)。
- [46] J. May, 《代数拓扑学简明教程》, 芝加哥数学讲义 (University of Chicago Press,
- Ward identities, Phys. Lett. B **37**, 95 (1971).
- [8] C. G. Callan, Jr. and J. A. Harvey, Anomalies and Fermion Zero Modes on Strings and Domain Walls, Nucl. Phys. B **250**, 427 (1985).
- [9] M. F. Atiyah, V. K. Patodi, and I. M. Singer, Spectral asymmetry and Riemannian Geometry 1, Math. Proc. Cambridge Phil. Soc. **77**, 43 (1975).
- [10] E. Witten, An SU(2) Anomaly, Phys. Lett. B **117**, 324 (1982).
- [11] E. Witten, GLOBAL GRAVITATIONAL ANOMALIES, Commun. Math. Phys. **100**, 197 (1985).
- [12] E. Witten and K. Yonekura, Anomaly Inflow and the η -Invariant, in *The Shoucheng Zhang Memorial Workshop* (2019) arXiv:1909.08775 [hep-th].
- [13] X.-G. Wen, Classifying gauge anomalies through symmetry-protected trivial orders and classifying gravitational anomalies through topological orders, Phys. Rev. D **88**, 045013 (2013), arXiv:1303.1803 [hep-th].
- [14] A. Kapustin and R. Thorngren, Higher Symmetry and Gapped Phases of Gauge Theories, Prog. Math. **324**, 177 (2017), arXiv:1309.4721 [hep-th].
- [15] D. S. Freed, Anomalies and Invertible Field Theories, Proc. Symp. Pure Math. **88**, 25 (2014), arXiv:1404.7224 [hep-th].
- [16] Y. Tachikawa, On gauging finite subgroups, SciPost Phys. **8**, 015 (2020), arXiv:1712.09542 [hep-th].
- [17] L. Bhardwaj and Y. Tachikawa, On finite symmetries and their gauging in two dimensions, JHEP **03**, 189, arXiv:1704.02330 [hep-th].
- [18] D. Gaiotto, A. Kapustin, N. Seiberg, and B. Willett, Generalized Global Symmetries, JHEP **02**, 172, arXiv:1412.5148 [hep-th].
- [19] J. McGreevy, Generalized Symmetries in Condensed Matter, Ann. Rev. Condensed Matter Phys. **14**, 57 (2023), arXiv:2204.03045 [cond-mat.str-el].
- [20] S. Schafer-Nameki, ICTP lectures on (non-)invertible generalized symmetries, Phys. Rept. **1063**, 1 (2024), arXiv:2305.18296 [hep-th].
- [21] L. Bhardwaj, L. E. Bottini, L. Fraser-Taliente, L. Glad-den, D. S. W. Gould, A. Platschorre, and H. Tillim, Lectures on generalized symmetries, Phys. Rept. **1051**, 1 (2024), arXiv:2307.07547 [hep-th].
- [22] R. Luo, Q.-R. Wang, and Y.-N. Wang, Lecture notes on generalized symmetries and applications, Phys. Rept. **1065**, 1 (2024), arXiv:2307.09215 [hep-th].
- [23] S.-H. Shao, What’s Done Cannot Be Undone: TASI Lectures on Non-Invertible Symmetries, arXiv (2023), arXiv:2308.00747 [hep-th].
- [24] X. Chen, Z.-C. Gu, Z.-X. Liu, and X.-G. Wen, Symmetry protected topological orders and the group cohomology of their symmetry group, Phys. Rev. B **87**, 155114 (2013), arXiv:1106.4772 [cond-mat.str-el].
- [25] A. Kapustin and R. Thorngren, Anomalies of discrete symmetries in three dimensions and group cohomology, Phys. Rev. Lett. **112**, 231602 (2014), arXiv:1403.0617 [hep-th].
- [26] A. Kapustin and R. Thorngren, Anomalies of discrete symmetries in various dimensions and group cohomology, arXiv (2014), arXiv:1404.3230 [hep-th].
- [27] D. V. Else and C. Nayak, Classifying symmetry-protected topological phases through the anomalous action of the symmetry on the edge, Phys. Rev. B **90**, 235137 (2014), arXiv:1409.5436 [cond-mat.str-el].
- [28] G. W. Moore and N. Seiberg, Classical and Quantum Conformal Field Theory, Commun. Math. Phys. **123**, 177 (1989).
- [29] R. Dijkgraaf and E. Witten, Topological Gauge Theories and Group Cohomology, Commun. Math. Phys. **129**, 393 (1990).
- [30] C.-M. Chang, Y.-H. Lin, S.-H. Shao, Y. Wang, and X. Yin, Topological Defect Lines and Renormalization Group Flows in Two Dimensions, JHEP **01**, 026, arXiv:1802.04445 [hep-th].
- [31] J. Milnor, On the homology of lie groups made discrete, Commentarii Mathematicae Helvetici **58**, 72 (1983).
- [32] The same 3-cocycle expression was derived in [48] for bosonic 2D theories using the method of Else and Nayak [27]. A recent discussion of $U(1)$ anomalies in the fermionic case within this framework can be found in [50].
- [33] Q. Jia, R. Luo, J. Tian, Y.-N. Wang, and Y. Zhang, Categorical Continuous Symmetry, arXiv (2025), arXiv:2509.13170 [hep-th].
- [34] D. S. Freed, M. J. Hopkins, J. Lurie, and C. Teleman, Topological Quantum Field Theories from Compact Lie Groups, in *A Celebration of Raoul Bott’s Legacy in Mathematics* (2009) arXiv:0905.0731 [math.AT].
- [35] A. Borel, Topology of lie groups and characteristic classes, Bulletin of the American Mathematical Society **61**, 397 (1955).
- [36] For some compact Lie groups, such as $U(1)$ and $SO(3)$, defining the Chern–Simons invariant at level $k = 1$ requires a spin structure [29, 51]. In these cases, one replaces the Chern–Simons term by the corresponding η -invariant (or equivalently, a quadratic refinement of the gauge fields pairing). This corresponds to *one half* of the generator of $H^4(BG, \mathbb{Z})$ and therefore can not be directly obtained as a class in $H^4(BG, \mathbb{Z})$. In this letter, we only require the orientability of the spacetime manifold and its extension throughout.
- [37] See more details on BG^δ in Appendix C. For a discussion of the simplicial complex construction of BG for finite G , the reader may consult section 9.1 in the lecture note <https://member.ipmu.jp/yuji.tachikawa/lectures/2024-mathphys/notes.pdf>.
- [38] We use the convention $\omega_{2n-1} := i^n \frac{(n-1)!}{(2n-1)!} \text{tr}(\tilde{g}^{-1} d\tilde{g})^{2n-1}$, such that it is normalized to integrate to 1 on the generator of $\pi_{2n-1}(G)$, and for $n = 2$, the generator of $\pi_3(G)$ is sent to the generator of $H_3(G, \mathbb{Z})$ [43].
- [39] S. P. Novikov, The Hamiltonian formalism and a many valued analog of Morse theory, Usp. Mat. Nauk **37N5**, 3 (1982).
- [40] S.-S. Chern and J. Simons, Characteristic forms and geometric invariants, Annals Math. **99**, 48 (1974).
- [41] E. Witten, On Holomorphic factorization of WZW and coset models, Commun. Math. Phys. **144**, 189 (1992).
- [42] D. S. Freed, Pions and Generalized Cohomology, J. Diff. Geom. **80**, 45 (2008), arXiv:hep-th/0607134.
- [43] Y. Lee, K. Ohmori, and Y. Tachikawa, Revisiting Wess-Zumino-Witten terms, SciPost Phys. **10**, 061 (2021), arXiv:2009.00033 [hep-th].
- [44] J. C. Baez and A. D. Lauda, Higher-Dimensional Algebra V: 2-Groups, arXiv (2003), arXiv:math/0307200.
- [45] S. Morita and A. M. Society, *Geometry of characteristic classes*, Iwanami series in modern mathematics (American Mathematical Society, 2001).
- [46] J. May, *A Concise Course in Algebraic Topology*, Chicago Lectures in Mathematics (University of Chicago Press,

- 1999).
- [47] 实际上, 可以将尚未明确指定的 w 定义为从 $H^3(\mathfrak{g}, \mathbb{R})$ 到 $H^3(B\bar{G}, \mathbb{R})$ 的同态, 使得 $\bar{f}^* \circ (kw|_{\bar{U}_k}) = \omega_3|_{\bar{U}_k}$ 在每个局部片 $\bar{U}_k$ 上成立。
- [48] K. Kawagoe 和 M. Levin, 《玻色对称性保护拓扑边界理论中的反常: 与 F 符号的联系及一种计算方法》, Phys. Rev. B 104, 115156 (2021). arXiv:2105.02909 [cond-mat.str-el]
- [49] H. W. Lenstra, 《原有限群》, 未发表讲义/预印本 (2003), <https://websites.math.leidenuniv.nl/algebra/Lenstra-Profinite.pdf>
- [50] M. Okada, S. Shimamura, Y. Tachikawa 和 Y. Zhang, 《论反常与费米子么正算符》, arXiv (2025), arXiv:2509.02989 [hep-th]
- [51] D. Belov 和 G. W. Moore, 《阿贝尔自旋 Chern-Simons 理论的分类》, arXiv (2005), arXiv:hep-th/0505235
- [52] J. D. Stasheff, 《群和分类空间的连续上调调》, Bulletin of the American Mathematical Society 84, 513 (1978)。
- [53] J. L. Dupont, 《曲率与特征类》, 数学讲义, 卷 640 (Springer, 1978)。
- [54] F. Kamber 和 P. Tondeur, 《群表示的平坦丛与特征类》, American Journal of Mathematics 89, 857 (1967)
- [55] P. Putrov, Q/Z 对称性, arXiv arXiv:2208.12071 [hep-th]
- [56] P. J. Hilton 和 U. Stambach, 《同调代数教程》, 数学研究生教材, 卷 4 (Springer, 纽约, 1971)。
- [57] A. Hatcher, 《代数拓扑学》, 剑桥大学出版社, 2002
- [58] E. Wofsey, 《Eilenberg-MacLane 空间的上同调 $k(\mathbb{R}, 1)$ 》, Math.StackExchange 对“Eilenberg-MacLane 空间的上同调 $K(\mathbb{R}, 1)$ ”的回答 (2019), 回答于2019年12月5日。
- [59] E. M. Friedlander 和 G. Mislin, 复李群及相关离散群的分类空间的上同调, 《Commentarii Mathematicae Helvetici》59, 347 (1984)

附录 A: $U(1)$ 全局对称性的 't Hooft 反常与李群上调调

正如我们已经指出的, 由于 $H^3(\mathfrak{g}, \mathbb{R}) = 0$, $U(1)$ 的 't Hooft 反常需要特别谨慎处理。F-移动由图 5 所示的移动给出。我们注意到, 这里的关键在于, 对于阿贝尔全局对称性, 必须

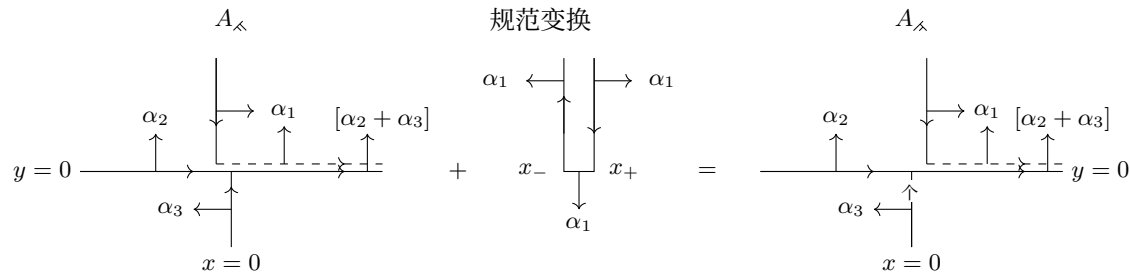


图 5. $U(1)$ 全局对称性的 F-移动。

考虑参数周期性的影响。因此, $e^{i\alpha_2}$ 与 $e^{i\alpha_3}$ 的融合结果是 $e^{i[\alpha_2+\alpha_3]}$, 而不是直观上的 $e^{i(\alpha_2+\alpha_3)}$ 。初始网络的庞加莱对偶为:

$$A_{\times} = i(\alpha_1\delta(x-x_-)H(y) - \alpha_3\delta(x)H(-y))dx + i(\alpha_1H(x-x_-) + \alpha_2H(-x) + [\alpha_2 + \alpha_3]H(x))\delta(y)dy \quad (\text{A1})$$

其中 $0 < \alpha_i < 1$ 。场强为 ($[\alpha] \equiv \alpha \bmod 2\pi\mathbb{Z}$):

$$F_{\times} = dA_{\times} = i([\alpha_2 + \alpha_3] - \alpha_2 - \alpha_3)\delta(x)\delta(y)dx \wedge dy. \quad (\text{A2})$$

对于规范变换, 我们有:

$$A_{\times} - A_{\times} = i\alpha_1(-\delta(x-x_-)H(y)dx - B(x; x_-, x_+)\delta(y)dy + \delta(x-x_+)H(y)dx) = d(-i\alpha_1B(x; x_-, x_+)H(y)) = d\epsilon. \quad (\text{A3})$$

因此, 我们有:

$$\epsilon = -i\alpha_1B(x; x_-, x_+)H(y). \quad (\text{A4})$$

可以计算得到反常为:

$$\begin{aligned} \mathcal{A}[A_{\times}; \epsilon] &= \frac{1}{(2\pi)^2} \int_{M_2} \epsilon dA_{\times} = \frac{1}{(2\pi)^2} \int_{M_2} \alpha_1(\alpha_2 + \alpha_3 - [\alpha_2 + \alpha_3])B(x; x_-, x_+)H(y)\delta(x)\delta(y)dx \wedge dy \\ &= \frac{1}{(2\pi)^2} \alpha_1(\alpha_2 + \alpha_3 - [\alpha_2 + \alpha_3]). \end{aligned} \quad (\text{A5})$$

- 1999).
- [47] Actually, one can define w , which has not yet been specified explicitly, to be the homomorphism from $H^3(\mathfrak{g}, \mathbb{R})$ to $H^3(B\bar{G}, \mathbb{R})$ such that $\bar{f}^* \circ (kw|_{\bar{U}_k}) = \omega_3|_{\bar{U}_k}$ holds on each local patch $\bar{U}_k$.
- [48] K. Kawagoe and M. Levin, Anomalies in bosonic symmetry-protected topological edge theories: Connection to F symbols and a method of calculation, Phys. Rev. B **104**, 115156 (2021), arXiv:2105.02909 [cond-mat.str-el].
- [49] H. W. Lenstra, Profinite groups, unpublished lecture notes / preprint (2003), available at <https://websites.math.leidenuniv.nl/algebra/Lenstra-Profinite.pdf>.
- [50] M. Okada, S. Shimamura, Y. Tachikawa, and Y. Zhang, On anomalies and fermionic unitary operators, arXiv (2025), arXiv:2509.02989 [hep-th].
- [51] D. Belov and G. W. Moore, Classification of Abelian spin Chern-Simons theories, arXiv (2005), arXiv:hep-th/0505235.
- [52] J. D. Stasheff, Continuous cohomology of groups and classifying spaces, Bulletin of the American Mathematical Society **84**, 513 (1978).
- [53] J. L. Dupont, *Curvature and Characteristic Classes*, Lecture Notes in Mathematics, Vol. 640 (Springer, 1978).
- [54] F. Kamber and P. Tondeur, Flat bundles and characteristic classes of group-representations, American Journal of Mathematics **89**, 857 (1967).
- [55] P. Putrov, Q/Z symmetry, arXiv (2022), arXiv:2208.12071 [hep-th].
- [56] P. J. Hilton and U. Stambach, *A Course in Homological Algebra*, Graduate Texts in Mathematics, Vol. 4 (Springer, New York, NY, 1971).
- [57] A. Hatcher, *Algebraic Topology*, Algebraic Topology (Cambridge University Press, 2002).
- [58] E. Wofsey, “cohomology of the eilenberg-maclane space $k(\mathbb{R}, 1)$ ”, Math.StackExchange answer to “Cohomology of the Eilenberg-MacLane space $K(\mathbb{R}, 1)$ ” (2019), answered Dec. 5, 2019.
- [59] E. M. Friedlander and G. Mislin, Cohomology of classifying spaces of complex lie groups and related discrete groups, Commentarii Mathematicae Helvetici **59**, 347 (1984).

Appendix A: 't Hooft Anomaly of $U(1)$ Global Symmetry and Lie Group Cohomology

As we have noted already, the 't Hooft anomaly of $U(1)$ needs extra care since $H^3(\mathfrak{g}, \mathbb{R}) = 0$. The F-move is given by the move shown in Figure 5. We note that here the key point is that for abelian global symmetry one has to

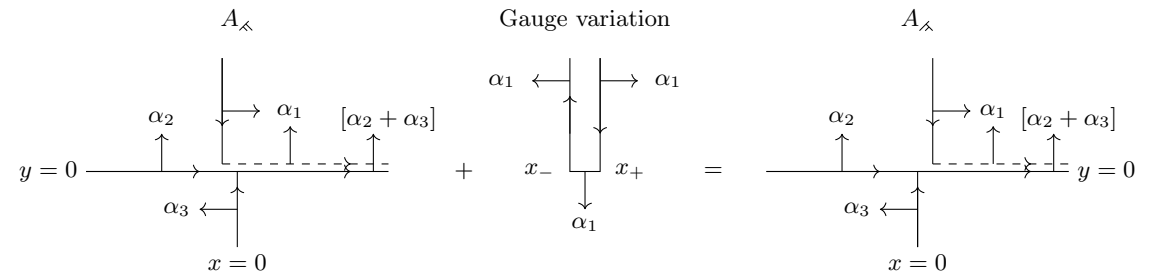


FIG. 5. The F-move for $U(1)$ global symmetry.

consider the consequence of the periodicity of the parameter. Thus the fusion of $e^{i\alpha_2}$ and $e^{i\alpha_3}$ results in $e^{i[\alpha_2+\alpha_3]}$ rather than naively $e^{i(\alpha_2+\alpha_3)}$. The Poincaré dual of the initial web is:

$$A_{\times} = i(\alpha_1\delta(x-x_-)H(y) - \alpha_3\delta(x)H(-y))dx + i(\alpha_1H(x-x_-) + \alpha_2H(-x) + [\alpha_2 + \alpha_3]H(x))\delta(y)dy \quad (\text{A1})$$

where $0 < \alpha_i < 1$. The field strength is ($[\alpha] \equiv \alpha \bmod 2\pi\mathbb{Z}$):

$$F_{\times} = dA_{\times} = i([\alpha_2 + \alpha_3] - \alpha_2 - \alpha_3)\delta(x)\delta(y)dx \wedge dy. \quad (\text{A2})$$

For the gauge variation we have:

$$A_{\times} - A_{\times} = i\alpha_1(-\delta(x-x_-)H(y)dx - B(x; x_-, x_+)\delta(y)dy + \delta(x-x_+)H(y)dx) = d(-i\alpha_1B(x; x_-, x_+)H(y)) = d\epsilon. \quad (\text{A3})$$

Therefore, we have:

$$\epsilon = -i\alpha_1B(x; x_-, x_+)H(y). \quad (\text{A4})$$

One can compute the anomaly to be:

$$\begin{aligned} \mathcal{A}[A_{\times}; \epsilon] &= \frac{1}{(2\pi)^2} \int_{M_2} \epsilon dA_{\times} = \frac{1}{(2\pi)^2} \int_{M_2} \alpha_1(\alpha_2 + \alpha_3 - [\alpha_2 + \alpha_3])B(x; x_-, x_+)H(y)\delta(x)\delta(y)dx \wedge dy \\ &= \frac{1}{(2\pi)^2} \alpha_1(\alpha_2 + \alpha_3 - [\alpha_2 + \alpha_3]). \end{aligned} \quad (\text{A5})$$

不难看出，上述结果推广了文献 [28] 中离散全局对称性的结果。

为说明在此情形下反常由李群上同调 $H^3\left(U(1)^\delta, U(1)\right)$ 描述，而不是由李代数上同调（后者实际上是平凡的）描述，我们回顾李群上同调的余边界算子 δ 定义为：

$$(\delta f)(g_1, \dots, g_{n+1}) = f(g_2, \dots, g_{n+1}) \times \prod_i f(g_1, \dots, g_i g_{i+1}, \dots, g_{n+1})^{(-1)^i} \times f(g_1, \dots, g_n) \quad (\text{A6})$$

对于 $f(e^{i\alpha_1}, e^{i\alpha_2}, e^{i\alpha_3}) = e^{i\alpha_1(\alpha_2+\alpha_3-[\alpha_2+\alpha_3])/(2\pi)}$ ，我们有：

$$\begin{aligned} f(e^{i\alpha_2}, e^{i\alpha_3}, e^{i\alpha_4}) &= e^{i\alpha_2(\alpha_3+\alpha_4-[\alpha_3+\alpha_4])/(2\pi)}, \\ f(e^{i(\alpha_1+\alpha_2)}, e^{i\alpha_3}, e^{i\alpha_4}) &= e^{i[\alpha_1+\alpha_2](\alpha_3+\alpha_4-[\alpha_3+\alpha_4])/(2\pi)}, \\ f(e^{i\alpha_1}, e^{i(\alpha_2+\alpha_3)}, e^{i\alpha_4}) &= e^{i\alpha_1([\alpha_2+\alpha_3]+\alpha_4-[[\alpha_2+\alpha_3]+\alpha_4])/(2\pi)}, \\ f(e^{i\alpha_1}, e^{i\alpha_2}, e^{i(\alpha_3+\alpha_4)}) &= e^{i\alpha_1(\alpha_2+[\alpha_3+\alpha_4]-[\alpha_2+[\alpha_3+\alpha_4]])/(2\pi)}, \\ f(e^{i\alpha_1}, e^{i\alpha_2}, e^{i\alpha_3}) &= e^{i\alpha_1(\alpha_2+\alpha_3-[\alpha_2+\alpha_3])/(2\pi)}, \end{aligned} \quad (\text{A7})$$

我们定义 $n_{ij} := \alpha_i + \alpha_j - [\alpha_i + \alpha_j] \in \{0, 2\pi\}$ 。因此我们有：

$$\begin{aligned} &(\delta f)(e^{i\alpha_1}, e^{i\alpha_2}, e^{i\alpha_3}, e^{i\alpha_4}) \\ &= \frac{f(e^{i\alpha_2}, e^{i\alpha_3}, e^{i\alpha_4})f(e^{i\alpha_1}, e^{i(\alpha_2+\alpha_3)}, e^{i\alpha_4})f(e^{i\alpha_1}, e^{i\alpha_2}, e^{i\alpha_3})}{f(e^{i(\alpha_1+\alpha_2)}, e^{i\alpha_3}, e^{i\alpha_4})f(e^{i\alpha_1}, e^{i\alpha_2}, e^{i(\alpha_3+\alpha_4)})} \\ &= e^{i(\alpha_2 n_{34} - (\alpha_1 + \alpha_2 - n_{12})n_{34} + \alpha_1(\alpha_2 + \alpha_3 + \alpha_4 - n_{23} - [\alpha_2 + \alpha_3 + \alpha_4]) - \alpha_1(\alpha_2 + \alpha_3 + \alpha_4 - n_{34} - [\alpha_2 + \alpha_3 + \alpha_4]) + \alpha_1 n_{23})/(2\pi)} \\ &= e^{in_{12}n_{34}/(2\pi)} = 1. \end{aligned} \quad (\text{A8})$$

因此，我们证明了 $f(\alpha_1, \alpha_2, \alpha_3) = e^{i\alpha_1(\alpha_2+\alpha_3-[\alpha_2+\alpha_3])/(2\pi)}$ 是关于 δ 的余循环。由于 f 和 f^k 都不是 δ -恰当的 [33,48]，我们证明了反常 $e^{2\pi i A[A_{\mathcal{A}}; \Lambda]} = e^{i\alpha_1(\alpha_2+\alpha_3-[\alpha_2+\alpha_3])/(2\pi)}$ 属于子群 $H^4(BU(1), \mathbb{Z}) \cong \mathbb{Z} \subset H^3(U(1)^\delta, U(1))$ 。

附录 B：高维群上同调与李代数上同调

我们还给出了次数为 $n > 3$ 的群上同调与李代数上同调的代表元，它们与 $(n-1)$ 时空维度中的反常相关。

对于阿贝尔群 $U(1)$ ，具有离散拓扑 $\mathbb{Z} \subset H^{2k+1}(U(1)^\delta, U(1))$ 的 $U(1)$ 群上同调子群的代表元为

$$f(e^{i\alpha_1}, e^{i\alpha_2}, \dots, e^{i\alpha_{2k+1}}) = e^{(2\pi)^{-k} i\alpha_1 \prod_{i=1}^k (\alpha_{2i} + \alpha_{2i+1} - [\alpha_{2i} + \alpha_{2i+1}])}. \quad (\text{B1})$$

可以直接验证闭性条件

$$\begin{aligned} (\delta f)(e^{i\alpha_1}, \dots, e^{i\alpha_{2k+2}}) &= f(e^{i\alpha_2}, \dots, e^{i\alpha_{2k+2}}) \cdot \left(\prod_{i=1}^{2k+1} f(e^{i\alpha_1}, \dots, e^{i[\alpha_i + \alpha_{i+1}]}, \dots, e^{i\alpha_{2k+1}})^{(-1)^i} \right) \cdot f(e^{i\alpha_1}, \dots, e^{i\alpha_{2k+1}}) \\ &= \exp \left(i(2\pi)^{-k} \prod_{i=1}^{k+1} (\alpha_{2i-1} + \alpha_{2i} - [\alpha_{2i-1} + \alpha_{2i}]) \right) \\ &= 1. \end{aligned} \quad (\text{B2})$$

我们还讨论了混合反常的群上同调代表元。对于二维中 $U(1)_1 \times U(1)_2$ 之间的混合反常，由规范不变的 4-形式反常多项式 $(2\pi)^{-2} F^{(1)} \wedge F^{(2)}$ 描述，它对应于 3-上闭链生成元 $f \in \mathbb{Z} \subset H^3(U(1)_1^\delta \times U(1)_2^\delta, U(1))$ ：

$$f((e^{i\alpha_1}, e^{i\beta_1}), (e^{i\alpha_2}, e^{i\beta_2}), (e^{i\alpha_3}, e^{i\beta_3})) = \exp(i\alpha_1(\beta_2 + \beta_3 - [\beta_2 + \beta_3])/(2\pi)). \quad (\text{B3})$$

其中 $e^{i\alpha_m} \in U(1)_1$ 和 $e^{i\beta_m} \in U(1)_2$ ($m = 1, 2, 3$)

。

One immediately recognizes that the above result generalizes the result for discrete global symmetry in [28].

To see that in this case the anomaly is described by the Lie group cohomology $H^3(U(1)^\delta, U(1))$ rather than by the Lie algebra cohomology (which is actually trivial), we recall that the coboundary operator δ for Lie group cohomology is defined as:

$$(\delta f)(g_1, \dots, g_{n+1}) = f(g_2, \dots, g_{n+1}) \times \prod_i f(g_1, \dots, g_i g_{i+1}, \dots, g_{n+1})^{(-1)^i} \times f(g_1, \dots, g_n) \quad (\text{A6})$$

For $f(e^{i\alpha_1}, e^{i\alpha_2}, e^{i\alpha_3}) = e^{i\alpha_1(\alpha_2+\alpha_3-[\alpha_2+\alpha_3])/(2\pi)}$ we have:

$$\begin{aligned} f(e^{i\alpha_2}, e^{i\alpha_3}, e^{i\alpha_4}) &= e^{i\alpha_2(\alpha_3+\alpha_4-[\alpha_3+\alpha_4])/(2\pi)}, \\ f(e^{i(\alpha_1+\alpha_2)}, e^{i\alpha_3}, e^{i\alpha_4}) &= e^{i[\alpha_1+\alpha_2](\alpha_3+\alpha_4-[\alpha_3+\alpha_4])/(2\pi)}, \\ f(e^{i\alpha_1}, e^{i(\alpha_2+\alpha_3)}, e^{i\alpha_4}) &= e^{i\alpha_1([\alpha_2+\alpha_3]+\alpha_4-[[\alpha_2+\alpha_3]+\alpha_4])/(2\pi)}, \\ f(e^{i\alpha_1}, e^{i\alpha_2}, e^{i(\alpha_3+\alpha_4)}) &= e^{i\alpha_1(\alpha_2+[\alpha_3+\alpha_4]-[\alpha_2+[\alpha_3+\alpha_4]])/(2\pi)}, \\ f(e^{i\alpha_1}, e^{i\alpha_2}, e^{i\alpha_3}) &= e^{i\alpha_1(\alpha_2+\alpha_3-[\alpha_2+\alpha_3])/(2\pi)}, \end{aligned} \quad (\text{A7})$$

We define $n_{ij} := \alpha_i + \alpha_j - [\alpha_i + \alpha_j] \in \{0, 2\pi\}$. Hence we have:

$$\begin{aligned} &(\delta f)(e^{i\alpha_1}, e^{i\alpha_2}, e^{i\alpha_3}, e^{i\alpha_4}) \\ &= \frac{f(e^{i\alpha_2}, e^{i\alpha_3}, e^{i\alpha_4})f(e^{i\alpha_1}, e^{i(\alpha_2+\alpha_3)}, e^{i\alpha_4})f(e^{i\alpha_1}, e^{i\alpha_2}, e^{i\alpha_3})}{f(e^{i(\alpha_1+\alpha_2)}, e^{i\alpha_3}, e^{i\alpha_4})f(e^{i\alpha_1}, e^{i\alpha_2}, e^{i(\alpha_3+\alpha_4)})} \\ &= e^{i(\alpha_2 n_{34} - (\alpha_1 + \alpha_2 - n_{12})n_{34} + \alpha_1(\alpha_2 + \alpha_3 + \alpha_4 - n_{23} - [\alpha_2 + \alpha_3 + \alpha_4]) - \alpha_1(\alpha_2 + \alpha_3 + \alpha_4 - n_{34} - [\alpha_2 + \alpha_3 + \alpha_4]) + \alpha_1 n_{23})/(2\pi)} \\ &= e^{in_{12}n_{34}/(2\pi)} = 1. \end{aligned} \quad (\text{A8})$$

Therefore we have proved that $f(\alpha_1, \alpha_2, \alpha_3) = e^{i\alpha_1(\alpha_2+\alpha_3-[\alpha_2+\alpha_3])/(2\pi)}$ is a cocycle with respect to δ . Since f both and f^k are not δ -exact [33, 48], we have proved that the anomaly $e^{2\pi i A[A_{\mathcal{A}}; \Lambda]} = e^{i\alpha_1(\alpha_2+\alpha_3-[\alpha_2+\alpha_3])/(2\pi)}$ is in the subgroup $H^4(BU(1), \mathbb{Z}) \cong \mathbb{Z} \subset H^3(U(1)^\delta, U(1))$.

Appendix B: Group and Lie algebra cohomology in higher dimensions

We also present the representatives for group and Lie algebra cohomologies with degree $n > 3$, which are relevant for anomalies in $(n-1)$ spacetime dimensions.

For abelian group $U(1)$, the representative for the subgroup of the $U(1)$ group cohomology with discrete topology $\mathbb{Z} \subset H^{2k+1}(U(1)^\delta, U(1))$ is given by

$$f(e^{i\alpha_1}, e^{i\alpha_2}, \dots, e^{i\alpha_{2k+1}}) = e^{(2\pi)^{-k} i\alpha_1 \prod_{i=1}^k (\alpha_{2i} + \alpha_{2i+1} - [\alpha_{2i} + \alpha_{2i+1}])}. \quad (\text{B1})$$

One can explicitly check the closedness condition

$$\begin{aligned} (\delta f)(e^{i\alpha_1}, \dots, e^{i\alpha_{2k+2}}) &= f(e^{i\alpha_2}, \dots, e^{i\alpha_{2k+2}}) \cdot \left(\prod_{i=1}^{2k+1} f(e^{i\alpha_1}, \dots, e^{i[\alpha_i + \alpha_{i+1}]}, \dots, e^{i\alpha_{2k+1}})^{(-1)^i} \right) \cdot f(e^{i\alpha_1}, \dots, e^{i\alpha_{2k+1}}) \\ &= \exp \left(i(2\pi)^{-k} \prod_{i=1}^{k+1} (\alpha_{2i-1} + \alpha_{2i} - [\alpha_{2i-1} + \alpha_{2i}]) \right) \\ &= 1. \end{aligned} \quad (\text{B2})$$

We also discuss the group cohomology representatives for mixed anomalies. For the mixed anomaly between $U(1)_1 \times U(1)_2$ in 2D, described by the gauge invariant 4-form anomaly polynomial $(2\pi)^{-2} F^{(1)} \wedge F^{(2)}$, it corresponds to the 3-cocycle generator $f \in \mathbb{Z} \subset H^3(U(1)_1^\delta \times U(1)_2^\delta, U(1))$:

$$f((e^{i\alpha_1}, e^{i\beta_1}), (e^{i\alpha_2}, e^{i\beta_2}), (e^{i\alpha_3}, e^{i\beta_3})) = \exp(i\alpha_1(\beta_2 + \beta_3 - [\beta_2 + \beta_3])/(2\pi)). \quad (\text{B3})$$

Here $e^{i\alpha_m} \in U(1)_1$ and $e^{i\beta_m} \in U(1)_2$ ($m = 1, 2, 3$).

更一般地，在 $2k$ 维时空中，对于 $U(1)_1 \times \cdots \times U(1)_l$ 之间的混合反常，由规范不变的 $(2k+2)$ -形式反常多项式 $(2\pi)^{-k-1} (F^{(1)})^{m_1} \cdots (F^{(l)})^{m_l}, m_1 + \cdots + m_l = k+1$ 描述，所提出的群上调生成元 $f \in \mathbb{Z} \subset H^{2k+1}(U(1)_1 \times \cdots \times U(1)_l, U(1))$ 为

$$\begin{aligned} & f((e^{i\alpha_1^{(1)}}, \dots, e^{i\alpha_1^{(l)}}), \dots, (e^{i\alpha_{2k+1}^{(1)}}, \dots, e^{i\alpha_{2k+1}^{(l)}})) \\ &= \exp(i(2\pi)^{-k} \alpha_1^{(1)} \left(\prod_{m=2}^{m_1} (\alpha_{2m-2}^{(1)} + \alpha_{2m-1}^{(1)} - [\alpha_{2m-2}^{(1)} + \alpha_{2m-1}^{(1)}]) \right) \\ & \cdot \prod_{j=2}^l \prod_{m=1}^{m_l} (\alpha_{2m_1+\cdots+2m_{j-1}+2m-2}^{(j)} + \alpha_{2m_1+\cdots+2m_{j-1}+2m-1}^{(j)} - [\alpha_{2m_1+\cdots+2m_{j-1}+2m-2}^{(j)} + \alpha_{2m_1+\cdots+2m_{j-1}+2m-1}^{(j)}]))). \end{aligned} \quad (\text{B4})$$

同样， $e^{i\alpha_m^{(j)}} \in U(1)_j, (m=1, \dots, 2k+1)$ 。

附录 C：反常、上调群与次级不变量。

a. F-变换与群上调调。对于有限群 G ，F-变换诱导拓扑网络的局部变换，不同构型相差一个相位。对于二维玻色理论，该相位因子由群上调调 $H^3(BG, \mathbb{R}/\mathbb{Z})$ 分类（我们将 $U(1) = \mathbb{R}/\mathbb{Z}$ 视为加法群），而在费米情形，相位因子由满足 $\delta\alpha = (-1)^{\nu^2}$ 的三元组 $(\mu, \nu, \alpha) \in Z^1(BG, \mathbb{Z}_2) \times Z^2(BG, \mathbb{Z}_2) \times C^3(BG, \mathbb{R}/\mathbb{Z})$ 决定。区别在于理论是否依赖自旋结构。在连续群框架下，我们只考虑玻色情形。有限群的另一个关键特征是，代数定义的上调群 $H^\bullet(G, A)$ 与分类空间 BG 的拓扑定义上调群 $H^\bullet(BG, A)$ 一致。

对于李群 G ， $H^\bullet(BG, A)$ 和 “ $H^\bullet(G, A)$ ” 都可以定义。前者通过考察分类空间，以与有限群情形相同的方式定义。然而，后者有多种版本。例如，可以用群上的多变量函数来定义；随后可将注意力限制于相对于群 G 的标准拓扑连续的函数，这个群通常记作带有下标表示连续性的 $H_c^\bullet(G, A)$ [52]。可以将 G^δ 取为带离散拓扑的群 G ，从而定义 $H_c^\bullet(G^\delta, A)$ ，它是有限群情形中 $H^\bullet(G, A)$ 的对应物。然而， $H_c^\bullet(G, A)$ 、 $H_c^\bullet(G^\delta, A)$ 和 $H_c^\bullet(G^\delta, A)$ 都不同于 $H^\bullet(BG, A)$ 。例如，当 G 紧致、连通且单连通时，对于 $n > 0$ ，有 $H_c^n(G, \mathbb{R}) = H_c^n(G, \mathbb{R}/\mathbb{Z}) = 0$ 。

由于任意平坦 G -丛都是 G^δ -丛，我们在平坦规范化情形下着重讨论 G^δ 。目的是说明存在一个类 $[\Omega] \in H_c^\bullet(G^\delta, \mathbb{R}/\mathbb{Z})$ ，它充当陈-西蒙斯不变量的对应物，并且就是反常。

b. 分类空间与次级不变量。给定平坦联络的模空间为 $\text{Hom}(\pi_1(M), G)/G$ ，可以验证 [45]，平坦 G -丛的分类空间 BG^δ 是如下的 Eilenberg–MacLane 空间

$$BG^\delta \cong K(G, 1). \quad (\text{C1})$$

赋予离散拓扑后， G^δ 的行为如同离散群，而 $H^\bullet(K(G, 1), A) \cong H^\bullet(BG^\delta, A) \cong H_c^\bullet(G^\delta, A)$ 则是拓扑定义的。注意，根据 [53] 的推论 (3.22)，平坦 G -丛有等价的定义

1. G -丛允许平坦联络；
2. G -丛允许一组常值过渡函数；
3. G -丛通过典范映射（实际上是恒等映射）将结构群约化为 G^δ

$$\iota: G^\delta \longrightarrow G. \quad (\text{C2})$$

正是第二种定义使我们能够通过拓扑缺陷网络描述平坦 G -丛。第三种定义也可改述为：存在一个 G^δ -丛，其在 ι_* 下的像（由 ι 诱导的 G^δ -丛与 G -丛之间的自然映射）是该平坦 G -丛 [54]。典范映射 ι 还由于 $B-$ 的函子性，在分类空间层面诱导出一个映射

$$B\iota: BG^\delta \longrightarrow BG. \quad (\text{C3})$$

More generally in $2k$ -spacetime dimensions, for a mixed anomaly between $U(1)_1 \times \cdots \times U(1)_l$, described by a gauge invariant $(2k+2)$ -form anomaly polynomial $(2\pi)^{-k-1} (F^{(1)})^{m_1} \cdots (F^{(l)})^{m_l}, m_1 + \cdots + m_l = k+1$, the proposed group cohomology generator $f \in \mathbb{Z} \subset H^{2k+1}(U(1)_1 \times \cdots \times U(1)_l, U(1))$ is

$$\begin{aligned} & f((e^{i\alpha_1^{(1)}}, \dots, e^{i\alpha_1^{(l)}}), \dots, (e^{i\alpha_{2k+1}^{(1)}}, \dots, e^{i\alpha_{2k+1}^{(l)}})) \\ &= \exp(i(2\pi)^{-k} \alpha_1^{(1)} \left(\prod_{m=2}^{m_1} (\alpha_{2m-2}^{(1)} + \alpha_{2m-1}^{(1)} - [\alpha_{2m-2}^{(1)} + \alpha_{2m-1}^{(1)}]) \right) \\ & \cdot \prod_{j=2}^l \prod_{m=1}^{m_l} (\alpha_{2m_1+\cdots+2m_{j-1}+2m-2}^{(j)} + \alpha_{2m_1+\cdots+2m_{j-1}+2m-1}^{(j)} - [\alpha_{2m_1+\cdots+2m_{j-1}+2m-2}^{(j)} + \alpha_{2m_1+\cdots+2m_{j-1}+2m-1}^{(j)}]))). \end{aligned} \quad (\text{B4})$$

Again $e^{i\alpha_m^{(j)}} \in U(1)_j, (m=1, \dots, 2k+1)$.

Appendix C: Anomalies, cohomology groups and secondary invariants.

a. *F-move and group cohomologies.* In the case of a finite group G , the F-move induces a local transformation of the topological network, with distinct configurations differing by a phase. For 2D bosonic theory, this phase factor is classified by the group cohomology $H^3(BG, \mathbb{R}/\mathbb{Z})$ (we treat $U(1) = \mathbb{R}/\mathbb{Z}$ as an additive group), while in the fermionic case the phase factor is determined by a triple $(\mu, \nu, \alpha) \in Z^1(BG, \mathbb{Z}_2) \times Z^2(BG, \mathbb{Z}_2) \times C^3(BG, \mathbb{R}/\mathbb{Z})$ such that $\delta\alpha = (-1)^{\nu^2}$. The distinction lies in whether the theory depends on spin structure. We will only consider the bosonic case in the continuous group setup. Another crucial feature of a finite group is that the algebraically defined cohomology group $H^\bullet(G, A)$ coincides with the topologically defined cohomology group $H^\bullet(BG, A)$ of the classifying space BG .

For a Lie group G , both $H^\bullet(BG, A)$ and “ $H^\bullet(G, A)$ ” can be defined. The former is defined the same way as in the finite group case by considering the classifying space. However, the latter have various versions. For example, it can be defined in terms of multivariable functions on the group; one can then restrict attention to the continuous functions with respect to the standard topology of the group G , this group is usually written as $H_c^\bullet(G, A)$ with the subscript indicating the continuousness [52]. One can take G^δ as the group G with discrete topology, in such a way we can define $H_c^\bullet(G^\delta, A)$, which is the analog of $H^\bullet(G, A)$ in the finite group case. However, both $H_c^\bullet(G, A)$ and $H_c^\bullet(G^\delta, A)$ differ from $H^\bullet(BG, A)$. For instance, if G is compact connected and simply connected, $H_c^n(G, \mathbb{R}) = H_c^n(G, \mathbb{R}/\mathbb{Z}) = 0$ for $n > 0$.

As any flat G -bundle is a G^δ -bundle, we focus on discussions of G^δ in the case of flat gauging. The aim is to show that there is a class $[\Omega] \in H_c^\bullet(G^\delta, \mathbb{R}/\mathbb{Z})$ that serves as an analog of the Chern–Simons invariant and it is the anomaly.

b. *Classifying space and secondary invariant.* Given the moduli space of flat connection is $\text{Hom}(\pi_1(M), G)/G$ one can verify [45] that the classifying space BG^δ for flat G -bundle is the following Eilenberg-MacLane space

$$BG^\delta \cong K(G, 1). \quad (\text{C1})$$

Equipped with discrete topology, G^δ behaves like a discrete group and $H^\bullet(K(G, 1), A) \cong H^\bullet(BG^\delta, A) \cong H_c^\bullet(G^\delta, A)$, where the formed is topologically defined. Note that one has equivalent definitions of flat G -bundle due to Corollary (3.22) of [53]

1. G -bundle admit flat connection;
2. G -bundle admit a set of constant transition functions;
3. G -bundle with a reduction of structure group to G^δ via the canonical map (actually the identity map)

$$\iota: G^\delta \longrightarrow G. \quad (\text{C2})$$

It is the second definition that allows one to describe a flat G -bundle by topological defect networks. The third definition can also be rephrased as, there exists a G^δ -bundle whose image under ι_* (the nature map between G^δ -bundles and G -bundles induced by ι) is the flat G -bundle [54]. The canonical map ι also induces a map at the level of classifying space due to the functoriality of $B-$

$$B\iota: BG^\delta \longrightarrow BG. \quad (\text{C3})$$

对于由 $\varphi : M \rightarrow BG$ 给出的任意平坦丛, 存在一个由 $\varphi^\delta : M \rightarrow BG^\delta$ 给出的 G^δ -丛, 使得下图交换

$$\begin{array}{ccc} & M & \\ \varphi^\delta \swarrow & & \searrow \varphi \\ BG^\delta & \xrightarrow{B\iota} & BG \end{array} \quad (C4)$$

有一个命题 ([53] 中的命题 9.1) 指出, 对于一般的李群 G , 复合映射为零

$$I(G) \xrightarrow{\text{Chern-Weil}} H^\bullet(BG, \mathbb{R}) \xrightarrow{B\iota^*} H^\bullet(BG^\delta, \mathbb{R}), \quad (C5)$$

其中 $I(G)$ 是不变多项式的集合。该命题告诉我们, $f \in I^n(G)$ 总是映射到 $0 \in H^{2n}(BG^\delta, \mathbb{R})$ 。

现在考虑以下交换的长正合序列图

$$\begin{array}{ccccccc} \dots & \xrightarrow{j^*} & H^{2n-1}(BG^\delta, \mathbb{R}/\mathbb{Z}) & \xrightarrow{\beta} & H^{2n}(BG^\delta, \mathbb{Z}) & \xrightarrow{i^*} & H^{2n}(BG^\delta, \mathbb{R}) \xrightarrow{j^*} \dots \\ & & \uparrow B\iota^* & & \uparrow B\iota^* & & \uparrow B\iota^* \\ \dots & \xrightarrow{j^*} & H^{2n-1}(BG, \mathbb{R}/\mathbb{Z}) & \xrightarrow{\beta} & H^{2n}(BG, \mathbb{Z}) & \xrightarrow{i^*} & H^{2n}(BG, \mathbb{R}) \xrightarrow{j^*} \dots, \end{array} \quad (C6)$$

其中垂直箭头只是各种系数的 $B\iota^*$, 而水平长正合序列由系数的标准短正合序列诱导

$$0 \longrightarrow \mathbb{Z} \xrightarrow{i} \mathbb{R} \xrightarrow{j} \mathbb{R}/\mathbb{Z} \longrightarrow 0, \quad (C7)$$

而 β 是连接同态, 通常也称为 Bockstein 同态。

· 如果 $[f] \in H^{2n}(BG, \mathbb{R})$ 是某个整系数类 $[f]_{\mathbb{Z}} \in H^{2n}(BG, \mathbb{Z})$ 的像, 即 $i^*([f]_{\mathbb{Z}}) = [f]$ 。由于 $B\iota^*([f]) = 0 \in H^{2n}(BG^\delta, \mathbb{R})$ 以及右侧方块的交换性推出 $i^*(B\iota^*([f]_{\mathbb{Z}})) = 0 \in H^{2n}(BG^\delta, \mathbb{R})$ 。

· 上方序列在 $H^{2n}(BG^\delta, \mathbb{Z})$ 处的正合性意味着存在一个类

$$[Tf] \in H^{2n-1}(BG^\delta, \mathbb{R}/\mathbb{Z}), \quad (C8)$$

使得 $\beta([Tf]) = B\iota^*([f]_{\mathbb{Z}})$, 其中 β 是上方长正合序列的Bockstein 同态。

总结上述讨论, 我们得到了一个典范映射

$$\begin{aligned} \kappa^\delta : H^{2n}(BG, \mathbb{Z}) &\longrightarrow H^{2n-1}(BG^\delta, \mathbb{R}/\mathbb{Z}) \\ [f]_{\mathbb{Z}} &\longmapsto [Tf]. \end{aligned} \quad (C9)$$

注意, 如果我们将 BG^δ 替换为 M , 并使用平坦的 G -丛 φ , 则得到零映射 (Chern-Weil $\circ \varphi^* = 0$) (现在复合映射为零, 因为曲率2-形式消失)

$$I(G) \xrightarrow{\text{Chern-Weil}} H^\bullet(BG, \mathbb{R}) \xrightarrow{\varphi^*} H^\bullet(M, \mathbb{R}), \quad (C10)$$

那么这将导出一个类 $CS \in H^{2n-1}(M, \mathbb{R}/\mathbb{Z})$, 即由Chern和Simons明确构造的Chern-Simons不变量; 我们刚刚通过图表追逐展示了如何以另一种方式得到它。因此, 固定一个平坦丛 φ 后, 我们还有另一个典范映射

$$\begin{aligned} \kappa : H^{2n}(M, \mathbb{Z}) &\longrightarrow H^{2n-1}(M, \mathbb{R}/\mathbb{Z}) \\ (c[f])_{\mathbb{Z}} &\longmapsto CS, \end{aligned} \quad (C11)$$

其中 $c([f])_{\mathbb{Z}}$ 是 $[f]_{\mathbb{Z}}$ 的整数特征类拉回到 M , 即 $(c[f])_{\mathbb{Z}} = \varphi^*([f]_{\mathbb{Z}})$ 。

将C4)、C9) 和C11) 结合起来, 我们得到交换图

$$\begin{array}{ccc} H^{2n}(BG, \mathbb{Z}) & \xrightarrow{\kappa^\delta} & H^{2n-1}(BG^\delta, \mathbb{R}/\mathbb{Z}) \\ \varphi^* \downarrow & & \downarrow (\varphi^\delta)^* \\ H^{2n}(M, \mathbb{Z}) & \xrightarrow{\kappa} & H^{2n-1}(M, \mathbb{R}/\mathbb{Z}), \end{array} \quad (C12)$$

For any flat bundle given by $\varphi : M \rightarrow BG$ there exists a G^δ -bundle by $\varphi^\delta : M \rightarrow BG^\delta$ such that the following diagram commutes

$$\begin{array}{ccc} & M & \\ \varphi^\delta \swarrow & & \searrow \varphi \\ BG^\delta & \xrightarrow{B\iota} & BG \end{array} \quad (C4)$$

There is a proposition (proposition 9.1 in [53]) stating that the composite map is zero for a general Lie group G

$$I(G) \xrightarrow{\text{Chern-Weil}} H^\bullet(BG, \mathbb{R}) \xrightarrow{B\iota^*} H^\bullet(BG^\delta, \mathbb{R}), \quad (C5)$$

where $I(G)$ is the set of invariant polynomials. This proposition tells us that $f \in I^n(G)$ is always mapped to $0 \in H^{2n}(BG^\delta, \mathbb{R})$.

We now consider the following commutative diagram of long exact sequences

$$\begin{array}{ccccccc} \dots & \xrightarrow{j^*} & H^{2n-1}(BG^\delta, \mathbb{R}/\mathbb{Z}) & \xrightarrow{\beta} & H^{2n}(BG^\delta, \mathbb{Z}) & \xrightarrow{i^*} & H^{2n}(BG^\delta, \mathbb{R}) \xrightarrow{j^*} \dots \\ & & \uparrow B\iota^* & & \uparrow B\iota^* & & \uparrow B\iota^* \\ \dots & \xrightarrow{j^*} & H^{2n-1}(BG, \mathbb{R}/\mathbb{Z}) & \xrightarrow{\beta} & H^{2n}(BG, \mathbb{Z}) & \xrightarrow{i^*} & H^{2n}(BG, \mathbb{R}) \xrightarrow{j^*} \dots, \end{array} \quad (C6)$$

where the vertical arrows are just $B\iota^*$ of various coefficients and the horizontal long exact sequences are induced by the standard short exact sequence of coefficients

$$0 \longrightarrow \mathbb{Z} \xrightarrow{i} \mathbb{R} \xrightarrow{j} \mathbb{R}/\mathbb{Z} \longrightarrow 0, \quad (C7)$$

and β is the connecting homomorphism, or called Bockstein homomorphism, conventionally.

• If $[f] \in H^{2n}(BG, \mathbb{R})$ is the image of some integral class $[f]_{\mathbb{Z}} \in H^{2n}(BG, \mathbb{Z})$, i. e. $i^*([f]_{\mathbb{Z}}) = [f]$. Because $B\iota^*([f]) = 0 \in H^{2n}(BG^\delta, \mathbb{R})$ and the commutativity of the right quadrat implies that $i^*(B\iota^*([f]_{\mathbb{Z}})) = 0 \in H^{2n}(BG^\delta, \mathbb{R})$.

• Exactness at the stage of $H^{2n}(BG^\delta, \mathbb{Z})$ in the upper sequence implies that there exists a class

$$[Tf] \in H^{2n-1}(BG^\delta, \mathbb{R}/\mathbb{Z}), \quad (C8)$$

such that $\beta([Tf]) = B\iota^*([f]_{\mathbb{Z}})$, where β is the Bockstein homomorphism of the upper long exact sequence.

Summarizing the above discussion, we found a canonical map

$$\begin{aligned} \kappa^\delta : H^{2n}(BG, \mathbb{Z}) &\longrightarrow H^{2n-1}(BG^\delta, \mathbb{R}/\mathbb{Z}) \\ [f]_{\mathbb{Z}} &\longmapsto [Tf]. \end{aligned} \quad (C9)$$

Note that if we replace BG^δ by M , and using a flat G -bundle φ get zero map (Chern-Weil $\circ \varphi^* = 0$) (now the composite map is zero because of the vanishing of the curvature 2-form)

$$I(G) \xrightarrow{\text{Chern-Weil}} H^\bullet(BG, \mathbb{R}) \xrightarrow{\varphi^*} H^\bullet(M, \mathbb{R}), \quad (C10)$$

then this would lead to a class $CS \in H^{2n-1}(M, \mathbb{R}/\mathbb{Z})$, namely Chern-Simons invariant constructed by Chern and Simons explicitly and we just demonstrated how one can obtain it alternatively via diagram chasing. Hence, fixing a flat bundle φ , we have another canonical map

$$\begin{aligned} \kappa : H^{2n}(M, \mathbb{Z}) &\longrightarrow H^{2n-1}(M, \mathbb{R}/\mathbb{Z}) \\ (c[f])_{\mathbb{Z}} &\longmapsto CS, \end{aligned} \quad (C11)$$

where $c([f])_{\mathbb{Z}}$ is the integral characteristic class of $[f]_{\mathbb{Z}}$ pulled back to M , i. e. $(c[f])_{\mathbb{Z}} = \varphi^*([f]_{\mathbb{Z}})$.

Combining (C4), (C9) and (C11) all together, we arrived at the commutative diagram

$$\begin{array}{ccc} H^{2n}(BG, \mathbb{Z}) & \xrightarrow{\kappa^\delta} & H^{2n-1}(BG^\delta, \mathbb{R}/\mathbb{Z}) \\ \varphi^* \downarrow & & \downarrow (\varphi^\delta)^* \\ H^{2n}(M, \mathbb{Z}) & \xrightarrow{\kappa} & H^{2n-1}(M, \mathbb{R}/\mathbb{Z}), \end{array} \quad (C12)$$

其中交换性由 $B-$ 的函子性推出, 并且有

$$(\varphi^\delta)^*([Tf]) = CS. \quad (\text{C13})$$

我们可以将其应用于异常的讨论。令 $n = 2$, 并例如取 $G = SU(2)$ 和 $c([f]_{\mathbb{Z}}) = c_2$ 为第二陈类。现在我们将 $[Tf]$ 简单记为 $[\Omega]$, 从而直接得到 $(\varphi^\delta)^*(\Omega) = \kappa(c_2) = CS$ 。

c. 迁移与同伦纤维。给定一个纤维化

$$F \longrightarrow E \xrightarrow{p} M, \quad (\text{C14})$$

迁移/悬挂[35]运算使我们能够将 M 的上同调类与 F 的上同调类联系起来, 其次数相差1。数学上, 这些细节依赖于 Leray-Serre谱序列。下面列出一些与我们的讨论相关的步骤

- 从一个上同调类 $[\alpha] \in H^{p+1}(M)$ 开始, 并假设其在 E 中的拉回使 $[p^*(\alpha)] = 0 \in H^{p+1}(E)$ 平凡化
- 则存在一个 p -上链, 记为 $\eta \in C^p(E)$, 使得 $\delta\eta = p^*(\alpha)$;
- 将 η 限制到纤维 F 上称为 $\overline{T\alpha}$ 和 $\delta(\overline{T\alpha}) = \underline{\underline{0}}$, 由此得到一个上同调类 $[\overline{T\alpha}] \in H^p(F)$, 而 $[\alpha]$ 称为 $[\overline{T\alpha}]$ 的传递。
- 一个主要例子是普适 G -丛

$$G \longrightarrow EG \xrightarrow{p} BG, \quad (\text{C15})$$

由于 EG 是可缩的, 其上同调是平凡的。因此, 通过上述过程我们得到悬挂映射

$$\tau : H^n(BG, A) \longrightarrow H_{\text{singular}}^{n-1}(G, A), \quad (\text{C16})$$

其中 A 是任意系数, 而 $x \in \text{Im}(\tau) \subset H_{\text{singular}}^{n-1}(G, A)$ 称为普适传递的。在以 $A = \mathbb{Z}$ 为系数且 $n = 4$ 的情况下, 映射 τ 一般不是满射, 但对于紧、连通、单连通且半单的李群, 它是同构（特别是对于每个紧、连通、单连通的单李群因子）。

我们所考虑的另一个重要纤维化如下引入。
按照同伦纤维的标准构造, 可以将映射 ι 写成一个纤维化

$$\overline{G} \longrightarrow E_\iota \xrightarrow{\iota} G, \quad (\text{C17})$$

其中 E_ι 定义为

$$E_\iota = \{(g, \gamma) \in G^\delta \times G^I \mid \gamma(0) = \iota(g)\} \subset G^\delta \times G^I, \quad (\text{C18})$$

它与 G^δ 同伦等价, 有时我们也直接写作 G^δ 。 G^I 是从区间 $I = [0, 1]$ 到 G 的连续映射空间, 是赋予紧开拓扑的拓扑空间。按构造, 同伦纤维 $\bar{G}$ 定义为

$$\overline{G} = \{(g, \gamma) \in G^\delta \times G^I \mid \gamma(0) = \iota(g), \gamma(1) = \iota(1_{G^\delta})\} \subset G^\delta \times G^I. \quad (\text{C19})$$

同伦纤维构造也可推广到 $B\iota$, 从而得到以下纤维化

$$B\bar{G} \longrightarrow BG^\delta \xrightarrow{B\iota} BG. \quad (\text{C20})$$

现在我们可以对纤维化 (C20) 应用悬挂运算, 并使用命题 (C5)。取 $f \in I^k(G)$ 使得 $[f] \in H^{2n}(BG, \mathbb{R})$, 则它确定一个类

$$[\overline{Tf}] \in H^{2n-1}(B\bar{G}, \mathbb{R}), \quad (\text{C21})$$

该类传递到 $[f]$ 。

in which the commutativity follows from the functoriality of $B-$ and we have

$$(\varphi^\delta)^*([Tf]) = CS. \quad (\text{C13})$$

We can apply this to our anomaly discussions. Let $n = 2$ and for instance take $G = SU(2)$ and $c([f]_{\mathbb{Z}}) = c_2$ the second Chern-class. We now denote $[Tf]$ simply by $[\Omega]$ and obtain straightforwardly $(\varphi^\delta)^*(\Omega) = \kappa(c_2) = CS$ 。

c. *Transgression and Homotopy fiber.* Given a fibration

$$F \longrightarrow E \xrightarrow{p} M, \quad (\text{C14})$$

the *trangression/suspension* [35] operation allows one to relate cohomology classes of M to cohomology classes of F with degree shifted by 1. Mathematically speaking, the details rely on the *Leray-Serre* spectral sequence. Here we list some steps relevant to our discussion

- One starts from a cohomology class $[\alpha] \in H^{p+1}(M)$ and assumes that its pullback in E trivializes $[p^*(\alpha)] = 0 \in H^{p+1}(E)$;
- Then there exists a p -cochain, denoted as $\eta \in C^p(E)$ such that $\delta\eta = p^*(\alpha)$;
- The restriction of η to the fiber F is called $\overline{T\alpha}$ and $\delta(\overline{T\alpha}) = 0$, one thus obtains a cohomology class $[\overline{T\alpha}] \in H^p(F)$ and $[\alpha]$ is called the *transgression* of $[\overline{T\alpha}]$.

A primary example is the universal G -bundle

$$G \longrightarrow EG \xrightarrow{p} BG, \quad (\text{C15})$$

where the cohomology of EG is trivial since it is contractible. Hence we have the *suspension* map by the above process

$$\tau : H^n(BG, A) \longrightarrow H_{\text{singular}}^{n-1}(G, A), \quad (\text{C16})$$

where A is an arbitrary coefficient and $x \in \text{Im}(\tau) \subset H_{\text{singular}}^{n-1}(G, A)$ is called *universal transgressive*. In the case of $A = \mathbb{Z}$ as coefficient and $n = 4$, the map τ is in general not surjective, but for compact, connected, simply connected and semi-simple Lie groups it is an isomorphism (in particular for each compact, connected, simply connected simple factor).

Another important fibration in our consideration can be introduced as follows.

Following the standard construction of *homotopy fiber*, one can write the map ι into a fibration

$$\overline{G} \longrightarrow E_\iota \xrightarrow{\iota} G, \quad (\text{C17})$$

where E_ι is given as

$$E_\iota = \{(g, \gamma) \in G^\delta \times G^I \mid \gamma(0) = \iota(g)\} \subset G^\delta \times G^I, \quad (\text{C18})$$

and it is homotopy equivalent to G^δ and we sometimes just write G^δ instead. G^I is the mapping space of continuous maps from the interval $I = [0, 1]$ to G , it is a topological space endowed with the compact-open topology. The homotopy fiber $\bar{G}$ is by construction defined as

$$\overline{G} = \{(g, \gamma) \in G^\delta \times G^I \mid \gamma(0) = \iota(g), \gamma(1) = \iota(1_{G^\delta})\} \subset G^\delta \times G^I. \quad (\text{C19})$$

The homotopy fiber construction also extends to $B\iota$ and we have the following fibration

$$B\bar{G} \longrightarrow BG^\delta \xrightarrow{B\iota} BG. \quad (\text{C20})$$

We can now apply the suspension operation to the fibration (C20) and use the proposition (C5). Pick $f \in I^k(G)$ such that $[f] \in H^{2n}(BG, \mathbb{R})$, then its determines a class

$$[\overline{Tf}] \in H^{2n-1}(B\bar{G}, \mathbb{R}), \quad (\text{C21})$$

that transgresses to $[f]$.

附录D：关于 $H^3\left(U(1)^\delta, U(1)\right)$

对 $\mathbb{Z}_N$ 的 “ $N \rightarrow \infty$ ” 极限进行数学上严谨的定义，需要使用直接极限（参见[55]，其中讨论了物理应用背景）。也就是说，在每当 n 整除 m 时，通过包含 $\mathbb{Z}_n \rightarrow \mathbb{Z}_m$ 构造一个直接系统，并取该系统的极限

$$\varinjlim \mathbb{Z}_n = \mathbb{Q}/\mathbb{Z}, \quad (\text{D1})$$

该极限的一个显式描述是取所有循环子群 $\mathbb{Z}_n \subset U(1)^\delta$ 的并，即

$$\mathbb{Q}/\mathbb{Z} = \bigcup_n \mathbb{Z}_n. \quad (\text{D2})$$

群 $\mathbb{Q}/\mathbb{Z}$ 包含 $U(1)^\delta$ 中的所有单位根，并且是 $U(1)^\delta$ 的挠子群。由于 $U(1)^\delta \cong \mathbb{R}^\delta/\mathbb{Z}$ ，我们有

$$U(1)^\delta \cong \mathbb{R}^\delta/\mathbb{Q} \oplus \mathbb{Q}/\mathbb{Z}. \quad (\text{D3})$$

上述同构源于 $\mathbb{Q}/\mathbb{Z}$ 是可除的，因此以下短正合序列分裂（尽管这不是规范的，并且取决于 $\mathbb{R}^\delta$ 的一组基的选择）

$$0 \longrightarrow \mathbb{Q}/\mathbb{Z} \longrightarrow \mathbb{R}^\delta/\mathbb{Z} \longrightarrow \mathbb{R}^\delta/\mathbb{Q} \longrightarrow 0. \quad (\text{D4})$$

承认选择公理， $\mathbb{R}^\delta$ 被视为具有不可数基的无限维 $\mathbb{Q}$ -向量空间。同样地，可除无挠群 $\mathbb{R}^\delta/\mathbb{Q}$ 也是具有不可数基的 $\mathbb{Q}$ -向量空间。设 $\mathbb{R}^\delta/\mathbb{Q}$ 的一个基的不可数指标集为 I ，则 $\mathbb{R}^\delta/\mathbb{Q} = \oplus^I \mathbb{Q}$ 。为简洁起见，我们也省略上标 δ ，因为下文不会引起混淆。我们将上述群记为 $A = \mathbb{R}^\delta/\mathbb{Q}$, $B = \mathbb{Q}/\mathbb{Z}$ 和 $G = \mathbb{R}^\delta/\mathbb{Z}$ 。在分类空间层面，我们处理的是 $K(G, 1) = K(A, 1) \times K(B, 1)$ 。我们可以应用万有系数定理

$$0 \longrightarrow \text{Ext}_{\mathbb{Z}}(H_2(G, \mathbb{Z}), U(1)) \longrightarrow H^3(G, U(1)) \cong H^3(K(G, 1), U(1)) \longrightarrow \text{Hom}_{\mathbb{Z}}(H_3(G, \mathbb{Z}), U(1)) \longrightarrow 0, \quad (\text{D5})$$

从而得到 $H^3(G, U(1)) \cong \text{Hom}_{\mathbb{Z}}(H_3(G, \mathbb{Z}), U(1))$ ，因为对于可除群 $U(1)$ ， $\text{Ext}_{\mathbb{Z}}(H_2(G, \mathbb{Z}), U(1))$ 消失。由于 $G = A \oplus B$ ，对于 $H_3(G, \mathbb{Z})$ ，我们有 Künneth 公式 [56]

$$0 \longrightarrow \bigoplus_{p+q=3} H_p(A, \mathbb{Z}) \otimes H_q(B, \mathbb{Z}) \longrightarrow H_3(G, \mathbb{Z}) \longrightarrow \bigoplus_{p+q=2} \text{Tor}(H_p(A, \mathbb{Z}), H_q(B, \mathbb{Z})) \longrightarrow 0, \quad (\text{D6})$$

并且该序列通过一个非自然分裂而分裂。现在回顾标准结果（可利用 [57] 第 3.F 节中的技术推导）

$$H_q(\mathbb{Q}/\mathbb{Z}, \mathbb{Z}) = \begin{cases} \mathbb{Z} & (q = 0) \\ \mathbb{Q}/\mathbb{Z} & (q \text{ odd}) \\ 0 & (q \text{ even and } q > 0) \end{cases}. \quad (\text{D7})$$

至于 $H_\bullet(A, \mathbb{Z})$ ，我们可以将其视为 Eilenberg–MacLane 空间 $K(A, 1)$ 的同调。于是，作为 Eilenberg–MacLane 空间的 $H_0(A, \mathbb{Z}) = H_0(K(A, 1), \mathbb{Z}) = \mathbb{Z}$ 是连通的，而 $H_1(A, \mathbb{Z}) = H_1(K(A, 1), \mathbb{Z}) = A$ 等于 $\pi_1(K(A, 1)) = A$ 在 A 下的阿贝尔化，因此是阿贝尔的。

对于挠积部分，我们需要计算 $\text{Tor}(H_0(A, \mathbb{Z}), H_2(B, \mathbb{Z}))$, $\text{Tor}(H_1(A, \mathbb{Z}), H_1(B, \mathbb{Z}))$ 和 $\text{Tor}(H_2(A, \mathbb{Z}), H_0(B, \mathbb{Z}))$ 。代入上述结果并利用 Tor 函子的性质，我们得到

$$\bigoplus_{p+q=2} \text{Tor}(H_p(A, \mathbb{Z}), H_q(B, \mathbb{Z})) = 0. \quad (\text{D8})$$

现在我们可以考虑 $(\overline{\text{D6}})$ 的张量积部分，为此我们需要两个额外成分 $H_2(A, \mathbb{Z})$ 和 $H_3(A, \mathbb{Z})$ ，它们出现在 $\mathbb{Q}/\mathbb{Z} \otimes H_2(A, \mathbb{Z})$ 和 $\mathbb{Z} \otimes H_3(A, \mathbb{Z}) = H_3(A, \mathbb{Z})$ 中。鉴于 $A = \oplus^I \mathbb{Q}$, $K(A, 1)$ 可被视为有限 n 上的 $K(\mathbb{Q}^n, 1)$ 的滤子同伦余极限，并且可以使用 Künneth 公式证明，对于所有 $p > 0$ ， $H_p(K(\mathbb{Q}^n, 1), \mathbb{Z})$ 都是一个 $\mathbb{Q}$ -向量空间。这对 $H_p(K(A, 1), \mathbb{Z})$ 同样成立，因为 $\mathbb{Q}$ -向量空间的滤子余极限仍是 $\mathbb{Q}$ -向量空间[58]。作为 $\mathbb{Q}$ -向量空间的 $H_2(A, \mathbb{Z})$ 是无挠的，因此

$$\mathbb{Q}/\mathbb{Z} \otimes H_2(A, \mathbb{Z}) = 0. \quad (\text{D9})$$

Appendix D: On $H^3(U(1)^\delta, U(1))$

A mathematically proper definition of the “ $N \rightarrow \infty$ ” limit of $\mathbb{Z}_N$ is via the *direct limit* (see [55] for a discussion in the context of physics applications). That is, build a direct system by the inclusion $\mathbb{Z}_n \rightarrow \mathbb{Z}_m$ whenever n divides m and take the limit of this system

$$\varinjlim \mathbb{Z}_n = \mathbb{Q}/\mathbb{Z}, \quad (\text{D1})$$

an explicit description of the limit is by taking the union of all cyclilc subgroups $\mathbb{Z}_n \subset U(1)^\delta$, i.e.

$$\mathbb{Q}/\mathbb{Z} = \bigcup_n \mathbb{Z}_n. \quad (\text{D2})$$

The group $\mathbb{Q}/\mathbb{Z}$ captures all roots of unity in $U(1)^\delta$ and it is the torsion subgroup of $U(1)^\delta$. Since $U(1)^\delta \cong \mathbb{R}^\delta/\mathbb{Z}$, we have

$$U(1)^\delta \cong \mathbb{R}^\delta/\mathbb{Q} \oplus \mathbb{Q}/\mathbb{Z}. \quad (\text{D3})$$

The above isomorphism is due to the fact that the $\mathbb{Q}/\mathbb{Z}$ is divisible and hence the following short exact sequence splits (although not canonical and depends on a choice of basis of $\mathbb{R}^\delta$)

$$0 \longrightarrow \mathbb{Q}/\mathbb{Z} \longrightarrow \mathbb{R}^\delta/\mathbb{Z} \longrightarrow \mathbb{R}^\delta/\mathbb{Q} \longrightarrow 0. \quad (\text{D4})$$

Admitting the axiom of choice, $\mathbb{R}^\delta$ is regarded as infinite dimensional $\mathbb{Q}$ -vector spaces with an uncountable basis. In the same vein, the divisible torsion-free group $\mathbb{R}^\delta/\mathbb{Q}$ is also a $\mathbb{Q}$ -vector spaces with a uncountable basis. Let us denote the uncountable index set of a basis of $\mathbb{R}^\delta/\mathbb{Q}$ by I then $\mathbb{R}^\delta/\mathbb{Q} = \oplus^I \mathbb{Q}$. We will also omit the superscript δ for brevity, as no confusion will arise in what follows. We denote the above groups by $A = \mathbb{R}^\delta/\mathbb{Q}$, $B = \mathbb{Q}/\mathbb{Z}$ and $G = \mathbb{R}^\delta/\mathbb{Z}$. At the level of classifying space, we are dealing with $K(G, 1) = K(A, 1) \times K(B, 1)$. We can apply the universal coefficient theorem

$$0 \longrightarrow \text{Ext}_{\mathbb{Z}}(H_2(G, \mathbb{Z}), U(1)) \longrightarrow H^3(G, U(1)) \cong H^3(K(G, 1), U(1)) \longrightarrow \text{Hom}_{\mathbb{Z}}(H_3(G, \mathbb{Z}), U(1)) \longrightarrow 0, \quad (\text{D5})$$

which yields $H^3(G, U(1)) \cong \text{Hom}_{\mathbb{Z}}(H_3(G, \mathbb{Z}), U(1))$ since $\text{Ext}_{\mathbb{Z}}(H_2(G, \mathbb{Z}), U(1))$ vanishes for the divisible group $U(1)$. As $G = A \oplus B$, we have for $H_3(G, \mathbb{Z})$ the Künneth formula [56]

$$0 \longrightarrow \bigoplus_{p+q=3} H_p(A, \mathbb{Z}) \otimes H_q(B, \mathbb{Z}) \longrightarrow H_3(G, \mathbb{Z}) \longrightarrow \bigoplus_{p+q=2} \text{Tor}(H_p(A, \mathbb{Z}), H_q(B, \mathbb{Z})) \longrightarrow 0, \quad (\text{D6})$$

and the sequence splits by an unnatural splitting. Now recall the standard result (which can be derived using the techniques in Section 3.F of [57])

$$H_q(\mathbb{Q}/\mathbb{Z}, \mathbb{Z}) = \begin{cases} \mathbb{Z} & (q = 0) \\ \mathbb{Q}/\mathbb{Z} & (q \text{ odd}) \\ 0 & (q \text{ even and } q > 0) \end{cases}. \quad (\text{D7})$$

As for $H_\bullet(A, \mathbb{Z})$, we can think of it as the homology of the Eilenberg-MacLane space $K(A, 1)$. Then $H_0(A, \mathbb{Z}) = H_0(K(A, 1), \mathbb{Z}) = \mathbb{Z}$ as Eilenberg-MacLane spaces are connected and $H_1(A, \mathbb{Z}) = H_1(K(A, 1), \mathbb{Z}) = A$ which equals the abelianization of $\pi_1(K(A, 1)) = A$ for A is abelian.

From the torsion product part we need to compute $\text{Tor}(H_0(A, \mathbb{Z}), H_2(B, \mathbb{Z}))$, $\text{Tor}(H_1(A, \mathbb{Z}), H_1(B, \mathbb{Z}))$ and $\text{Tor}(H_2(A, \mathbb{Z}), H_0(B, \mathbb{Z}))$. By inserting the above results and using the properties of the Tor functor we conclude

$$\bigoplus_{p+q=2} \text{Tor}(H_p(A, \mathbb{Z}), H_q(B, \mathbb{Z})) = 0. \quad (\text{D8})$$

Now we can consider the tensor product part of (D6), for this we need two extra ingredients $H_2(A, \mathbb{Z})$ and $H_3(A, \mathbb{Z})$ which appear in $\mathbb{Q}/\mathbb{Z} \otimes H_2(A, \mathbb{Z})$ and $\mathbb{Z} \otimes H_3(A, \mathbb{Z}) = H_3(A, \mathbb{Z})$. Given $A = \oplus^I \mathbb{Q}$, $K(A, 1)$ can be thought as filtered homotopy colimit of $K(\mathbb{Q}^n, 1)$ over finite n and one can use Künneth formula to show that $H_p(K(\mathbb{Q}^n, 1), \mathbb{Z})$ is a $\mathbb{Q}$ -vector space for all $p > 0$. This is also true for $H_p(K(A, 1), \mathbb{Z})$, as filtered colimit of $\mathbb{Q}$ -vector spaces are $\mathbb{Q}$ -vector spaces [58]. $H_2(A, \mathbb{Z})$ as a $\mathbb{Q}$ -vector spaces is torsion-free and hence

$$\mathbb{Q}/\mathbb{Z} \otimes H_2(A, \mathbb{Z}) = 0. \quad (\text{D9})$$

我们还剩下 $H_3(A, \mathbb{Z})$ 和 $H_0(A, \mathbb{Z}) \otimes H_3(B, \mathbb{Z}) = \mathbb{Z} \otimes \mathbb{Q}/\mathbb{Z} = \mathbb{Q}/\mathbb{Z}$ 。

总结上述计算，我们得到

$$H^3(G, U(1)) = \text{Hom}(H_3(A, \mathbb{Z}) \oplus \mathbb{Q}/\mathbb{Z}, U(1)) = H^3(\mathbb{R}^\delta/\mathbb{Q}, U(1)) \oplus \hat{\mathbb{Z}}, \quad (\text{D10})$$

其中 $\hat{\mathbb{Z}}$ 是赋范整数环，可以看作直积 $\prod_p R_p$ ，其中 R_p 是 p -进整数环。现在，通过将每个整数 n 映射到其模所有正整数的剩余类序列，可将 $\mathbb{Z}$ 规范嵌入 $\hat{\mathbb{Z}}$ 中。该映射由将 $U(1)$ 的3级 $2k$ 反常多项式（按我们的规范化为 k Chern-Simons不变量）嵌入 $H^3\left(U(1)^\delta, U(1)\right)$ 的 $\hat{\mathbb{Z}}$ 部分的3-余循环来刻画。

注意，在上述讨论中， $H_3(A, \mathbb{Z}) = H_3(K(A, 1), \mathbb{Z})$ 仍是一个未指定的 $\mathbb{Q}$ -向量空间，维数可能不可数。再加上 $H^\bullet(K(A, 1), \mathbb{Z})$ 的上同调环，这超出了我们目前的理解范围，我们希望数学家有朝一日能够阐明这些问题。

对于非阿贝尔李群，这个问题的复杂性急剧增加。一般来说，要准确确定上调群非常困难。我们将简要提及[31, 59]中的一些内容。设 G 为一个具有有限个连通分支的任意李群，则

- 由公式 (C3) 中的典范映射 $B\iota$ 诱导的映射 $B\iota^* : H^\bullet(BG, \mathbb{Z}) \rightarrow H^\bullet(BG^\delta, \mathbb{Z})$ 是单射；
- Friedlander-Milnor 猜想。典范映射 $B\iota$ 诱导出具有模 p 系数的同调与上调同构，或更一般地，具有任意有限系数群的同调与上调同构。

附录E：规范变换的反常

在本附录中，我们将回顾二维中的规范反常，并指出该讨论可自然地推广到更高维。设有一个定义在 M_2 上的二维理论，其反常由反常多项式 $I_4(F)$ 表征

$$I_4(F) = \frac{k}{2} \text{Tr} F \wedge F, \quad (\text{E1})$$

它是闭的，并且局部上是恰当的

$$I_4(F) = dI_3(A), \quad (\text{E2})$$

其中 $I_3(A) \equiv CS_k(A)$ 是局部陈-西蒙斯密度。考虑由 Λ 参数化的无穷小规范变换， $I_3(A)$ 的规范变分由于

$$d\delta_\Lambda I_3(A) = \delta_\Lambda dI_3(A) = \delta_\Lambda I_4(F) = 0, \quad (\text{E3})$$

也是局部恰当的。于是，配分函数 $Z(A)$ 的反常通过下降方程计算为

$$\delta_\Lambda I_3(A) = d\mathcal{A}[A; \Lambda], \quad (\text{E4})$$

其形式为

$$\delta_\Lambda \log Z[A] = 2\pi i \int_{M_2} \mathcal{A}[A; \Lambda]. \quad (\text{E5})$$

接下来考虑有限规范变换

$$A \rightarrow g^{-1} A g + g^{-1} d g, \quad (\text{E6})$$

该变换由 M_2 上的群函数 $g \equiv g(x, y)$ 参数化，并且与单位元连通。让我们构造一个满足 $t \in [0, 1]$ 的三维流形 $M_2 \times [0, 1]$ ，并将二维规范函数 g 延拓到 $g(t) \equiv g(x, y, t)$ 中，如下

$$g(x, y, 0) = e, \quad g(x, y, 1) = g(x, y). \quad (\text{E7})$$

我们还按照以下方式，将 M_2 上的规范场 A 沿 $[0, 1]$ 延拓

$$A(t) = g(t)^{-1} A_0 g(t) + g(t)^{-1} d g(t), \quad (\text{E8})$$

We are left with $H_3(A, \mathbb{Z})$ and $H_0(A, \mathbb{Z}) \otimes H_3(B, \mathbb{Z}) = \mathbb{Z} \otimes \mathbb{Q}/\mathbb{Z} = \mathbb{Q}/\mathbb{Z}$.

Summarized the above computation we arrive at

$$H^3(G, U(1)) = \text{Hom}(H_3(A, \mathbb{Z}) \oplus \mathbb{Q}/\mathbb{Z}, U(1)) = H^3(\mathbb{R}^\delta/\mathbb{Q}, U(1)) \oplus \hat{\mathbb{Z}}, \quad (\text{D10})$$

where $\hat{\mathbb{Z}}$ is the profinite integers and can be though as the direct product $\prod_p R_p$ where R_p is the ring of p -adic integer. Now there is a canonical embedding of $\mathbb{Z}$ into $\hat{\mathbb{Z}}$ by sending each integer n to the sequence of its residues modulo all positive integers. This map is modeled by embedding the level- k anomaly polynomial (level- $2k$ Chern-Simons invariant in our normalisation) of $U(1)$ into the 3-cocycle of $\hat{\mathbb{Z}}$ part of $H^3(U(1)^\delta, U(1))$.

Note that in the above consideration, $H_3(A, \mathbb{Z}) = H_3(K(A, 1), \mathbb{Z})$ remains an unspecified $\mathbb{Q}$ -vector space, possibly of uncountable dimension. Together with the cohomology ring of $H^\bullet(K(A, 1), \mathbb{Z})$, this lies beyond our current understanding, and we hope it will one day be clarified by mathematicians.

For non-abelian Lie groups, the complexity of this question increases dramatically. It is in general very hard to determine the cohomology group exactly. We will just briefly mention some contents from [31, 59]. Let G be an arbitrary Lie group with finitely many components, then

- the map $B\iota^* : H^\bullet(BG, \mathbb{Z}) \rightarrow H^\bullet(BG^\delta, \mathbb{Z})$ induced by the canonical map $B\iota$ from the Eq. (C3) is injective;
- *Friedlander-Milnor conjecture*. The canonical map $B\iota$ induces isomorphisms of homology and cohomology with mod p coefficients, or more generally with any finite coefficient group.

Appendix E: Anomaly of gauge transformation

In this appendix, we will review the gauge anomaly in 2D, noting that the discussion naturally extends to higher dimensions. Suppose we have a 2D theory living on M_2 whose anomaly is characterized by the anomaly polynomial $I_4(F)$

$$I_4(F) = \frac{k}{2} \text{Tr} F \wedge F, \quad (\text{E1})$$

which is closed and is locally exact

$$I_4(F) = dI_3(A), \quad (\text{E2})$$

where $I_3(A) \equiv CS_k(A)$ is the local Chern-Simons density. Consider an infinitesimal gauge transformation parametrized by Λ , the gauge variation of $I_3(A)$ is closed due to

$$d\delta_\Lambda I_3(A) = \delta_\Lambda dI_3(A) = \delta_\Lambda I_4(F) = 0, \quad (\text{E3})$$

and is also locally exact. Then the anomaly of the partition function $Z(A)$ is computed via the descent equation

$$\delta_\Lambda I_3(A) = d\mathcal{A}[A; \Lambda], \quad (\text{E4})$$

and takes the form

$$\delta_\Lambda \log Z[A] = 2\pi i \int_{M_2} \mathcal{A}[A; \Lambda]. \quad (\text{E5})$$

We then consider a finite gauge transformation

$$A \rightarrow g^{-1} A g + g^{-1} d g, \quad (\text{E6})$$

parametrized by a group function $g \equiv g(x, y)$ on M_2 and is connected to the identity. Let us construct a 3D manifold $M_2 \times [0, 1]$ with $t \in [0, 1]$ and extend the 2D gauge function g into $g(t) \equiv g(x, y, t)$ as

$$g(x, y, 0) = e, \quad g(x, y, 1) = g(x, y). \quad (\text{E7})$$

We also extend the gauge field A on M_2 along $[0, 1]$ according to

$$A(t) = g(t)^{-1} A_0 g(t) + g(t)^{-1} d g(t), \quad (\text{E8})$$

其中, 将 A_0 视为 M_2 上 A 的平凡延拓, 即 $A_0(x, y, t) = A(x, y)$ 。于是, 与单位元连通的有限规范变换 g 所产生的二维反常为

$$\delta_g \log Z[A] = 2\pi i \int_{[0,1] \times M_2} CS_k[A(t)]. \quad (\text{E9})$$

首先, 可以证明, 在固定边界条件 $g(0) = e, g(1) = g$ 的情况下, 上述反常与延拓 $g(t)$ 无关。原因在于, 只要固定边界条件, 陈-西蒙斯积分在规范变换下保持不变。其次, 当 $g(t)$ 很小时, 有

$$\int_{[0,1] \times M_2} CS[A(t)] = \int_{[0,1] \times M_2} CS[A(t)] - \int_{[0,1] \times M_2} CS[A_0] = \int_{[0,1] \times M_2} \delta_{g(t)} CS[A], \quad (\text{E10})$$

其中在中间步骤中我们使用了 $CS[A_0] = 0$, 因为 A_0 是平凡延拓。如上所示, 陈-西蒙斯密度的规范变分是一个全导数

$$\delta_{g(t)} CS(A) = d\mathcal{A}[A; \Lambda], \quad (\text{E11})$$

其中 $\Lambda(t)$ 是与 $e^{i\Lambda(t)} = g(t)$ 相关的无穷小规范参数。利用 $g(0) = e$ 和 $\Lambda(0) = 0$ 这一事实, 我们得到无穷小规范反常的形式 (E5)。

where A_0 is considered as a trivial extension of A on M_2 as $A_0(x, y, t) = A(x, y)$. The 2D anomaly of the finite gauge transformation g connected to the identity is then evaluated as

$$\delta_g \log Z[A] = 2\pi i \int_{[0,1] \times M_2} CS_k[A(t)]. \quad (\text{E9})$$

First, one can show that the anomaly given above is independent of the extension $g(t)$ with fixed boundary conditions $g(0) = e, g(1) = g$. The reason is that the Chern-Simons integral is invariant under the gauge transformation as long as we fix the boundary condition. Second, if $g(t)$ is small, we have

$$\int_{[0,1] \times M_2} CS[A(t)] = \int_{[0,1] \times M_2} CS[A(t)] - \int_{[0,1] \times M_2} CS[A_0] = \int_{[0,1] \times M_2} \delta_{g(t)} CS[A], \quad (\text{E10})$$

where in the middle we use $CS[A_0] = 0$ since A_0 is a trivial extension. As shown above, the gauge variation of Chern-Simons density is a total derivative

$$\delta_{g(t)} CS(A) = d\mathcal{A}[A; \Lambda], \quad (\text{E11})$$

where $\Lambda(t)$ is the infinitesimal gauge parameter related by $e^{i\Lambda(t)} = g(t)$. Using the fact $g(0) = e$ and $\Lambda(0) = 0$, we recover the infinitesimal version of the gauge anomaly (E5).